

**OCEANSENSE: AN IOT-BASED HYBRID SYSTEM FOR
REAL-TIME DETECTION AND TRACKING OF
MARINE DEBRIS**

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ENERGY-RESILIENT ADAPTIVE COMMUNICATION (ERAC) PROTOCOL

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Declaration of the Candidate and Supervisor

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Abstract

This study presents the Energy-Resilient Adaptive Communication (ERAC) protocol, a cross-layer Medium Access Control (MAC) framework designed for energy-constrained marine IoT sensor networks within the OceanSense system. These networks operate in dynamic ocean-surface environments where nodes rely on stochastic energy harvesting from solar and wave sources while maintaining long-range communication using LoRa. ERAC addresses key challenges such as fluctuating energy availability, intermittent connectivity due to wave-induced occlusion, and the need for priority-based data transmission. The protocol integrates predictive energy management using Exponentially Weighted Moving Average (EWMA) techniques and adaptive duty cycling to maintain Energy Neutral Operation (ENO). A Quality of Service (QoS)-aware scheduling mechanism prioritizes critical debris detection alerts over routine data, ensuring reliable communication under constrained conditions. Validation is performed through NS-3 simulations and hardware implementation using ESP32-based LoRa nodes. Results demonstrate improved packet delivery ratio, reduced latency for high-priority traffic, and extended network lifetime compared to non-adaptive MAC approaches. The findings highlight the importance of combining predictive energy awareness with QoS-driven scheduling to enable resilient and autonomous marine monitoring systems.

Keywords: Energy Harvesting, LoRa, MAC Protocol, Marine IoT, QoS

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List of Abbreviations

Abbreviation	Description
ACK	Acknowledgement
BMS	Battery Management System
ENO	Energy Neutral Operation
ERAC	Energy-Resilient Adaptive Communication
EWMA	Exponentially Weighted Moving Average
GPIO	General Purpose Input/Output
HAL	Hardware Abstraction Layer
I2C	Inter-Integrated Circuit
IMU	Inertial Measurement Unit
INA219	Current/Voltage Monitoring IC (Texas Instruments)
IoT	Internet of Things
ISM	Industrial, Scientific and Medical (radio band)
LoRa	Long Range (chirp-spread-spectrum modulation)
LoRaWAN	Long Range Wide Area Network
LPWAN	Low-Power Wide Area Network
MAC	Medium Access Control
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
NS-3	Network Simulator 3
PDR	Packet Delivery Ratio
QoS	Quality of Service
RPi	Raspberry Pi
RSSI	Received Signal Strength Indicator
SF	Spreading Factor (LoRa)
SLIIT	Sri Lanka Institute of Information Technology
SOC	State of Charge
SPI	Serial Peripheral Interface
STM32	STMicroelectronics 32-bit Microcontroller
TENG	Triboelectric Nanogenerator
UART	Universal Asynchronous Receiver/Transmitter
WSN	Wireless Sensor Network

Introduction

1.1 Executive Summary

The Energy-Resilient Adaptive Communication (ERAC) protocol is designed to enable long-lived marine IoT sensor networks by jointly managing unpredictable energy inputs and intermittent connectivity. In the ocean environment, renewable IoT buoys (sensing temperature, salinity, pollution, etc.) face harsh conditions: strong waves and weather cause volatile energy (from solar/wave harvesters) and block radio links, while critical data (e.g. pollution alerts) must be delivered with priority [1]. Existing literature shows that marine IoT is increasingly important for monitoring climate change and ocean health [1], and LoRa/LPWAN offers low-power long-range links for such buoys [2]. However, prior work has largely addressed these problems in isolation. For example, Buoy systems have been built using LoRaWAN with solar energy [2], and innovative harvesters (triboelectric generators) demonstrate wave power for continuous IoT buoys [3, 4], but no MAC-layer protocol yet integrates stochastic energy availability with priority-based traffic and wave-induced connectivity loss. ERAC fills this gap by: (a) using energy predictions (lightweight EWMA forecast) to adapt duty-cycling, (b) scheduling multi-level QoS priority queues so urgent ocean alerts are sent first, and (c) cross-layer optimization between MAC and routing to handle irregular link outages (wave occlusions) and energy state. We will evaluate ERAC via detailed NS-3 simulations (measuring latency, delivery, energy use) and a hardware prototype (e.g. ESP32+LoRa buoys). Major contributions include a predictive MAC for energy-neutral operation (ENO), QoS scheduling for marine events, and an open framework for simulation and prototyping. The work will include figures (system diagrams, protocol flowcharts) and empirical charts (e.g. research timeline, focus distribution). This introduction reports on relevant literature, gaps, and precise objectives for ERAC.

1.2 Background Literature

- **Marine IoT and Sensing Challenges:**

Ocean monitoring is vital (climate change, marine health) and is now increasingly enabled by IoT sensor networks [1, 5]. Conventional methods (ships, satellites) are costly and sparse; IoT buoys with wireless links (e.g. LoRa) allow persistent, high-resolution monitoring of parameters like temperature, salinity, pH, etc. [5, 1]. Xu et al. (2019) note that climate concerns have driven marine environment monitoring and that IoT (and Big Data) are transforming this field [5]. In IoT-based marine systems, devices must be **rugged and autonomous**: waterproof, mechanically robust against waves/typhoons, and energy-efficient [5]. Crucially, Xu et al. list **energy harvesting and radio stability** as key marine requirements [5]: nodes must harvest ambient energy (sun/waves) to offset long links, and buoy motion/weather can easily break line-of-sight radio paths [1, 6]. This motivates ERAC's focus on stochastic energy and connectivity.

- **Energy Harvesting for Marine Sensors:**

Ocean-surface sensors often use **hybrid harvesters**. Solar panels are common for buoys [2], but cloudy or stormy seas can cut solar power at critical times [2, 7]. Water waves contain abundant energy, especially in storms [7]. Linear “heaving buoy” generators have been shown to supply 1–10 W for ocean sensors [3], demonstrating feasibility. More recently, triboelectric–electromagnetic **nanogenerators** (OHNG) embedded in floating buoys have achieved omnidirectional wave capture [7]. Zhang et al. demonstrate a passive buoy that uses a hybrid TENG/EMG to capture unpredictable wave energy and power a LoRa node [7]. These works highlight that ambient energy in the marine environment is highly variable: wave and solar harvests can fluctuate by orders of magnitude [7]. Reviews of EH-IoT note that careful energy management (e.g. prediction and adaptive duty cycling) is essential for perpetual operation [8, 9].

- **LoRa/LPWAN for Marine Communication:**

Long-range, low-power radio (LoRa/LoRaWAN) is emerging as a practical choice for marine IoT [2, 10]. Majumder et al. built prototype buoys with sensors and LoRaWAN uplinks; solar PV and batteries power the nodes, and data is sent via a coastal gateway [2]. They demonstrate that LoRaWAN can support real-time meteorological monitoring from moored buoys with energy harvesting [2]. Similarly, a recent open-source buoy (OLB) using LoRa achieved **>2 km range and up to 7 months battery life** on lakes/fjords [10], showing LoRa's potential. However, LoRa links in open sea face unique impairments: Liu et al. observe that seawater and dynamic buoy motion severely degrade LoRa signal (e.g. up to 12% PRR loss at 1 m waves [6]). In fact, strong absorption by saltwater was noted to "severely degrade" LoRa comms, despite LoRa's promise for marine use [6]. In rough seas, experiments show Link PRR can drop below 88% [6]. To cope, recent systems schedule transmissions during lulls: one study found data is sent preferentially in "good" weather windows to avoid losses [6]. These insights motivate ERAC to adapt timing to channel conditions.

- **Energy-Aware MAC Protocols:**

In general, IoT/WSNs, many MAC-layer protocols aim to minimize idle listening and adapt to energy availability [11, 9]. A common goal is **energy-neutral operation (ENO)**: keep consumption below harvesting so nodes never die. For example, Sarang et al. (2011) propose an ENO-based adaptive duty cycle MAC where nodes stay in ENO as long as possible, using surplus harvest to boost performance [9]. Sakib et al. (2022) introduce EQPD-MAC, a cross-layer MAC that supports prioritized data with adaptive active periods [11]. EQPD-MAC uses an "adaptive active time" to limit idle listening, integrates routing, and handles multi-hop traffic; in simulations it cut node energy use by ~30% while improving throughput [11]. These works show that duty cycling and priority scheduling can greatly extend sensor lifetime in WSNs. However, most assume stable channels; ERAC must extend these ideas to an ocean context with intermittent links.

- **QoS-aware Scheduling:**

High-priority data (emergencies) must preempt routine sensing. QoS MAC protocols have been proposed for WSNs to meet such needs [11]. Sakib’s EQPD-MAC categorizes packets by urgency and gives critical alerts faster service [11]. But none of these directly account for marine link outages; their scheduling assumes transmissions can occur when scheduled. ERAC will build on this concept of priority queues but also incorporate channel awareness (e.g. delaying non-critical packets until link recovers).

- **Cross-layer Designs:**

Integrating MAC with networking (cross-layer) often yields better energy efficiency and reliability. Sandhiyaa and Gomathy (2023) describe an underwater sensor network where MAC slot scheduling (via reinforcement learning) is combined with energy-aware routing for QoS [12]. They show that coupling load-balanced MAC and hybrid routing improves performance in dynamic conditions [12]. ERAC similarly adopts a cross-layer view: MAC duty cycles will inform routing choices (and vice versa) so that critical data can be forwarded even under energy/channel constraints.

The table below summarizes key related works (proposed problems, methods, results, etc.) relevant to ERAC:

Table 1.2.1 - Summary of Related Work Relevant to ERAC

Reference	Year	Problem Addressed	Methods	Key Results	Relevance to ERAC
[1] Xu et al.	2019	IoT in marine environment	Literature review (IoT + Big Data)	Identifies architecture challenges: energy & connectivity	Highlights marine IoT demands
[11] Sakib et al.	2022	Energy-QoS MAC for WSNs	EQPD-MAC (adaptive duty + routing)	Cuts energy by ~30%, boosts throughput 23%	Example of prioritized energy-aware MAC

[6] Liu et al.	2025	LoRa reception in marine links	FCLoRa: dual-domain accumulation	PRR maintained at SNR -35 dB	Demonstrates LoRa marine challenges
[2] Majumder et al.	2024	LoRaWAN buoys for coastal data	Prototype buoys + LoRaWAN + solar	Real-time temp/salinity monitoring	Shows LoRaWAN + solar EH viability
[7] Zhang et al.	2025	Omnidirectional wave harvesting	Hybrid TENG/EMG nanogenerator	Efficient wave capture for passive buoy	Novel wave EH for marine IoT
[10] Dreyer et al.	2026	Open-source LoRa buoy (OLB)	LoRa drifter; field tests	>2 km range, ~7 months battery life	Validates long-range LoRa operation
[12] Sandhiyaa & Gomathy	2023	QoS & energy in UWSN	Cross-layer MAC (RL) + routing	Improved load-balancing and reliability	Example of cross-layer QoS approach
[9] Vigorito et al.	2007	Adaptive duty cycling	ENO-based adaptive MAC	Demonstrated ENO feasibility	ENO model baseline for ERAC

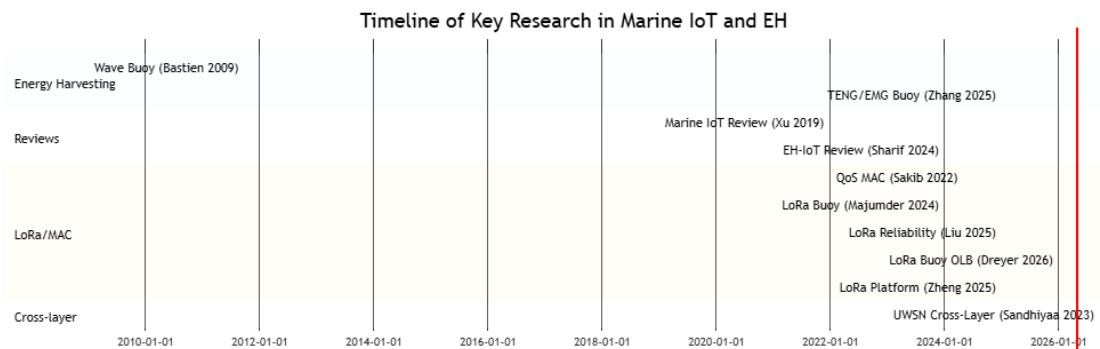


Figure 1.2.1 - Timeline of Key Research in Marine Io and Energy Harvesting

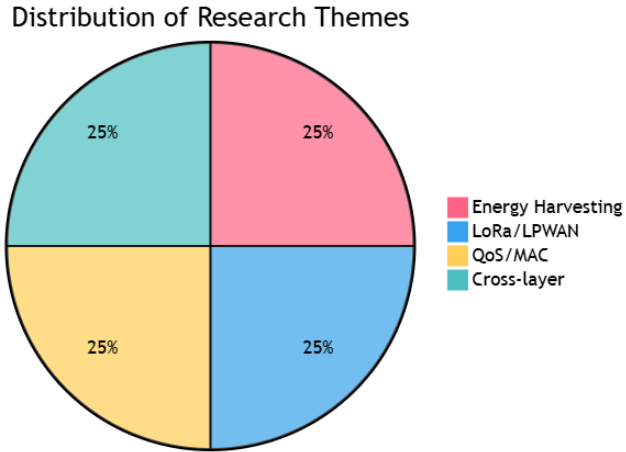


Figure 1.2.2 - Distribution of Research Themes

1.3 Research Gap

Despite this progress, **critical gaps remain**. No prior protocol explicitly ties together (i) **stochastic marine energy** (waves and solar that vary unpredictably) with (ii) **intermittent connectivity** (buoy radio occlusion from wave motion) and (iii) **priority-driven traffic** in a MAC-layer solution. Existing MAC and scheduling schemes either assume stable energy inputs or fixed connectivity [11, 9]. For example, Sakib et al. propose prioritized MACs for WSNs [11], but they do not account for energy harvesting or link breakage. Conversely, energy-neutral MACs [9] assume packets always eventually send. In the marine context, however, a buoy may have no harvest during storms and simultaneously face *signal blackout* from waves [7, 6]. In this case, urgent alerts (like oil spills) must preempt other data, but must wait for a brief connectivity window. The literature lacks a protocol that jointly adapts duty cycle and scheduling based on energy forecasts (e.g. using EWMA models [9]) and on-the-fly link quality, while enforcing multi-level QoS. In summary:

- **Energy vs Connectivity:** Wave energy harvest is bursty, and waves also sever radio links. How to remain energy-neutral and still guarantee critical data delivery under such dual randomness has not been addressed.
- **Priority Traffic under Uncertainty:** Marine IoT data includes heterogeneous priorities (safety alerts vs routine samples). Existing protocols either lack priority support or assume good connectivity [11].
- **Cross-layer Missing:** No marine-specific cross-layer design exists that couples MAC scheduling with routing in EH-WSNs. Underwater work [12] hints at this, but does not consider RF-based buoys or EH.

These gaps motivate ERAC: a MAC-layer protocol for marine buoys that is adaptive to both energy and channel states, supports prioritized queues, and is validated against real marine conditions.

1.4 Research Problem

The core **problem statement** is: Design and evaluate an adaptive MAC protocol for marine IoT buoys that ensures reliable, energy-neutral communications under intermittent LoRa connectivity and stochastic energy harvesting. In scope are ocean-surface sensor nodes with general hardware (e.g. microcontroller + LoRa transceiver, battery, solar panel and/or wave harvester), moored within $\sim 0\text{--}5$ km of a gateway (scale not fixed). Key assumptions: we do **not** specify a fixed deployment scale or exact sensors; instead, ERAC must work for generic node counts. We assume each node can estimate its own energy (battery/SOC) and link quality (via RSSI/PDR feedback). The **specific problem** is to schedule transmissions and sleep periods so that:

- The node's energy consumption never exceeds its harvest long-term (energy-neutral).
- High-priority packets (e.g. alarm signals) are sent as soon as possible, even if it means delaying normal traffic.

- Data is buffered when the link is down (due to wave occlusion) and sent when conditions improve. We will formulate this with an analytical model (e.g. Markov chain or optimization) and implement ERAC in NS-3 and on hardware.

1.5 Research Objectives and Approach

Primary Objective: Develop ERAC, an energy-resilient MAC protocol that achieves energy-neutral operation (ENO) while providing prioritized QoS for marine sensor data under intermittent LoRa links.

Specific Objectives:

1. **Energy Prediction & Duty Cycling:** Implement a lightweight EWMA-based energy forecasting model on each node that predicts near-future harvest. Adapt the node's duty cycle (sleep/awake period) dynamically so that average consumption stays at or below predicted harvest (thus achieving ENO) [9, 13].
2. **Priority Scheduling:** Introduce multi-level packet queues (e.g. emergency, standard, bulk). Design MAC scheduling rules so that higher-priority queues get earlier access in active periods, possibly with reserved mini-slots, while lower-priority packets wait. Ensure this is done **fairly** to avoid starvation.
3. **Link-Aware Operation:** Incorporate **link quality sensing** (e.g. RSSI, ACK rate) to detect occlusion. When link drops, ERAC will enter a buffered sleep cycle (still harvesting) and resume transmissions once conditions recover. Optionally, integrate weather forecasts or sea-state info if available.
4. **Cross-layer Integration:** Combine MAC with routing logic: if a parent node is energy-starved or out of contact, allow rerouting via neighbors (if multi-hop is used). Use information from MAC (e.g. duty patterns) to influence route choice for critical data.

Hypotheses:

- *H1*: Predictive duty-cycling will keep nodes in ENO, yielding lifetimes close to infinity (exceeding base MAC by >50%).
- *H2*: Priority scheduling will reduce end-to-end delay for urgent packets by X% (e.g. $\geq 20\%$) while only modestly affecting normal traffic.
- *H3*: Together, ERAC will improve delivery ratio of high-priority data in rough conditions relative to baseline MAC by mitigating losses through smart timing.

Proposed Approach and Contributions:

- We will design ERAC's algorithms (pseudo-code/flowcharts) and implement them as an NS-3 module. A novel EWMA predictor for wave/solar energy (tuned to marine data) will drive duty-cycle adaptation. Priority queues and backoff rules will be embedded in a LoRa MAC framework.
- **Contributions:** (a) ERAC protocol design; (b) open-source NS-3 model; (c) demonstration that ENO and QoS can coexist for marine IoT; (d) analysis of energy vs. latency trade-offs in this context.

Evaluation Methods:

- **Simulation:** Using NS-3 (with a LoRaWAN module), we will simulate various scenarios (node counts, energy availabilities, wave patterns). Metrics include node lifetime, packet delivery ratio (overall and per priority), latency of priority packets, and network throughput. Performance will be compared against standard LoRaWAN MAC and a static duty-cycling MAC.
- **Hardware Prototyping:** We will build a small testbed of 3–5 LoRa-enabled buoys (e.g. ESP32 with LoRa module, solar panel, battery, accelerometer) and a gateway. Real ocean tests (or a large wave tank) will measure ERAC's behavior under real wave motion (e.g. using [61†L2936-L2944] insights) and solar variation. We will instrument for energy (battery voltage) and packet stats.

Limitations and Ethical/Environmental Considerations:

- **Limitations:** ERAC will be validated for coastal buoy scales (km-range). Very large networks or deep underwater acoustics are out of scope. The hardware assumptions (solar panel + LoRa) may not hold for all cases. The protocol does not control physical movement or external factors (pirates, marine life).
- **Ethical/Environmental:** Ocean sensor deployments must be non-polluting. We will use biodegradable buoy materials where possible and ensure no interference with marine fauna (LoRa's low power is unlikely harmful). Data privacy is minimal as sensors measure environment, not personal info. Energy harvesting can reduce carbon footprint by avoiding frequent battery swaps. We will follow ethical research protocols for field tests (permits, safety at sea).

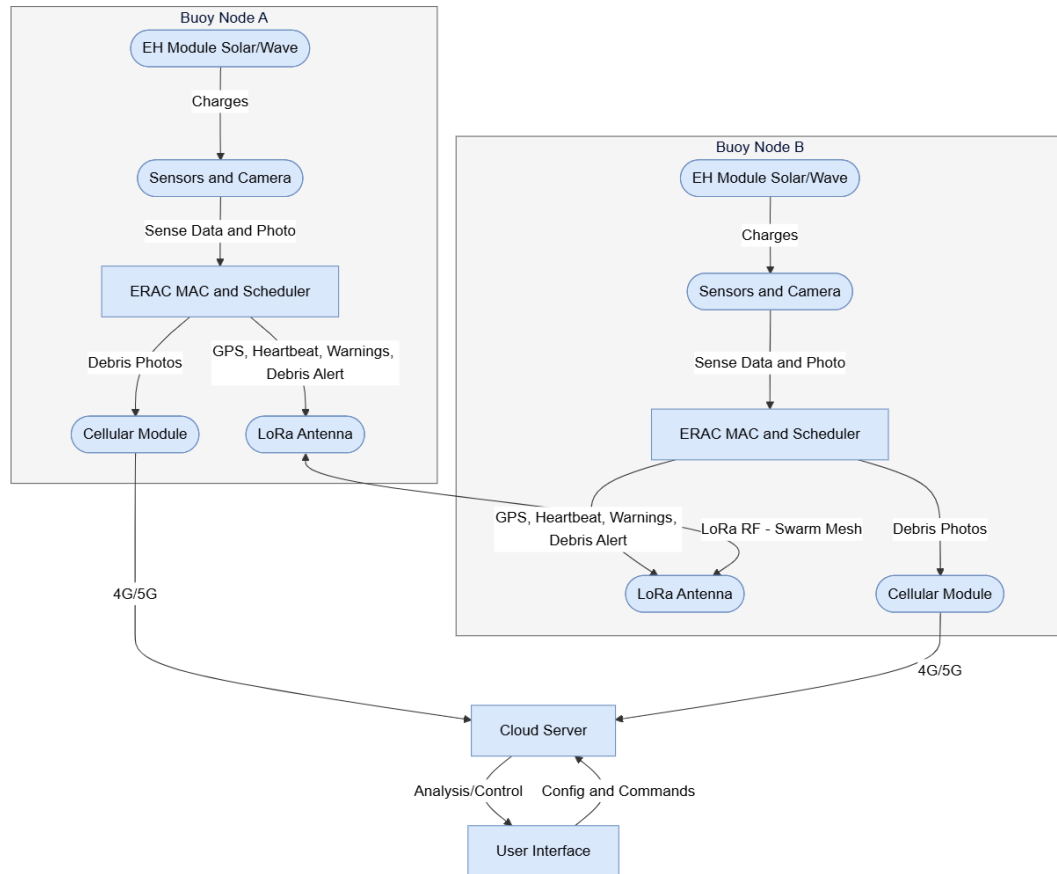


Figure 1.5.1 - ERAC-Based Marine IoT Buoy Node and LoRa Network Architecture

The above diagram shows the ERAC system: each buoy node integrates sensors, a camera, and an energy harvester feeding the ERAC MAC scheduler. Lightweight

metadata like GPS, heartbeat, warnings, and debris alerts are exchanged between nodes over LoRa within the swarm. Debris images are transmitted separately via cellular to the cloud server, where data is processed and made accessible through a user interface.

Methodology

2.1 Methodology Overview

The methodology adopted for the design, implementation, and evaluation of the Energy-Resilient Adaptive Communication (ERAC) protocol follows a structured, iterative engineering approach. The methodology was organized into five principal phases: (1) requirements analysis and protocol design, (2) firmware implementation on embedded hardware, (3) NS-3 network simulation, (4) hardware integration and physical testing, and (5) performance evaluation and analysis. Each phase produced concrete artefacts - specifications, firmware modules, simulation scripts, and measurement datasets - that fed into subsequent phases. This approach ensured that design decisions were grounded in literature evidence from Chapter 1, systematically validated through simulation before hardware commitment, and finally confirmed through physical deployment.

The overarching methodology is underpinned by three guiding engineering principles that shaped every design decision throughout the project:

- **Energy Neutrality First:** Any design decision that risks depleting a node's battery permanently is rejected in favour of a more conservative alternative, even at the cost of reduced throughput. The Energy Neutral Operation (ENO) constraint is treated as a hard constraint, not a soft optimisation target.
- **Priority-Driven Communication:** The protocol must never sacrifice high-priority packet delivery to conserve energy. Energy adaptation occurs by deferring or discarding lower-priority traffic, not by degrading the delivery of critical data.
- **Simplicity and Deployability:** Algorithms must execute efficiently on a resource-constrained microcontroller (STM32F401, 84 MHz, 512 KB flash, 96 KB SRAM) without requiring floating-point-heavy computation or large memory allocations. Complexity is bounded to ensure real-time operation under FreeRTOS task scheduling.

Figure 2.1 below illustrates the overall methodology pipeline, from requirements through to performance evaluation.

[Figure 2.1]

ERAC Research Methodology Pipeline: Requirements → Protocol Design → Firmware Implementation → NS-3 Simulation → Hardware Integration → Evaluation

Figure 2.1 – ERAC Research Methodology Pipeline

2.2 System Architecture and Design Approach

Before the ERAC protocol could be designed, a clear understanding of the target system architecture was necessary. ERAC operates as a middleware firmware layer within the broader OceanSense multi-node buoy network. This section describes the network topology, individual node architecture, and energy subsystem that constrain and motivate the protocol's design choices.

2.2.1 Network Topology and Deployment Model

The OceanSense system employs a star-of-stars network topology. A central coastal gateway node — typically a Raspberry Pi equipped with a LoRa module — serves as the data aggregation point. Multiple autonomous buoy nodes, each running the ERAC firmware, are deployed within LoRa range of the gateway (approximately 0.5 km to 5 km depending on sea conditions and antenna height). Nodes do not communicate directly with each other in the baseline topology; all uplink traffic is directed toward the gateway, which relays data to a cloud platform for further processing.

Each node operates independently, without synchronisation with other nodes. There is no global time reference or scheduled slot allocation imposed by the gateway. Instead, ERAC's adaptive duty cycling and probabilistic backoff mechanisms organically prevent persistent collision between nodes sharing the LoRa channel. The LoRa

channel operates in the 433 MHz or 868 MHz ISM band, depending on regional regulatory constraints.

[Figure 2.2]

OceanSense Network Topology: Star topology with coastal gateway at centre; 5–10 buoy nodes at varying distances; LoRa uplink arrows; wave/solar energy icons on each buoy

Figure 2.2 – OceanSense Network Topology and Deployment Model

2.2.2 Individual Node Architecture

Each buoy node consists of four hardware subsystems that interact under ERAC firmware control:

1. **Sensing Subsystem:** Responsible for acquiring environmental data. Sensors include a turbidity sensor and camera module for debris detection, a GPS module for node localization, and temperature and salinity probes for oceanographic monitoring. The sensing subsystem interfaces with the STM32 via UART and I2C buses.
2. **Energy Subsystem:** Provides power to all node components from harvested and stored energy. Described in detail in Section 2.2.3.
3. **Processing Subsystem:** The STM32F401CCU6 microcontroller runs FreeRTOS and hosts all ERAC firmware modules. A Raspberry Pi Zero 2W co-processor handles computationally intensive tasks such as image-based debris classification, powered on demand by ERAC's MOSFET-based power switching to avoid unnecessary idle drain.
4. **Communication Subsystem:** An SX1276-based LoRa module provides the long-range radio link to the gateway. The module is connected to the STM32 via SPI. ERAC controls all transmission timing, spreading factor selection, and power level settings through this interface.

[Figure 2.3]

Node Block Diagram: Four subsystems (Sensing / Energy / Processing / Communication) with internal bus connections and ERAC firmware layer highlighted as middleware

Figure 2.3 – Individual OceanSense Node Block Diagram

2.2.3 Energy Subsystem Design

The energy subsystem is the most critical hardware element from ERAC's perspective, as the protocol's entire decision logic depends on accurate knowledge of energy state. The subsystem was designed with the following components:

Table 2.2.1 - Energy Subsystem Components and Their Roles

Component	Specification	Role in ERAC
Solar Panel	5W, 18V monocrystalline	Primary daytime energy source
Wave Energy Harvester	Linear electromagnetic generator, ~0.5–3W	Supplementary harvest in wave conditions
Li-ion Battery Pack	2S2P, 7.4V nominal, 4400 mAh	Primary energy storage buffer
Battery Management System (BMS)	Overcharge, over-discharge, short-circuit protection	Battery safety and charge control
INA219 Sensor (×2)	12-bit ADC, ±3.2A / 32V	Real-time current and voltage measurement
MOSFET Switch	N-channel, logic-level gate	Controlled power gating for Raspberry Pi subsystem
5V Buck Converter	LM2596, up to 3A output	Regulated supply rail for STM32 and peripherals

Two INA219 sensors are deployed: one monitors the combined solar and wave harvest current (input rail), and the other monitors the system load current (output rail). The difference between these measurements, integrated over time, yields the net energy balance and allows ERAC to continuously update the battery state-of-charge (SOC) estimate. This dual-sensor arrangement is more robust than monitoring the battery voltage alone, which responds slowly to current changes due to battery internal resistance and hysteresis.

[Figure 2.4]

Energy Subsystem Architecture: Solar Panel → INA219 (harvest monitor) → BMS → Battery → INA219 (load monitor) → 5V Buck → STM32; MOSFET branch for RPi; Wave harvester feeds same input rail

Figure 2.4 – Energy Subsystem Architecture and Power Flow

2.3 ERAC Protocol Design

This section presents the formal design of the ERAC protocol, covering its energy model, state machine, priority scheduling logic, and cross-layer architecture. The design is the primary intellectual contribution of this individual work and translates the requirements identified in Chapter 1 into a concrete, implementable protocol specification.

2.3.1 Design Philosophy and Goals

ERAC is a cross-layer MAC protocol. Unlike classical layered designs — where the physical, MAC, network, and application layers exchange information only through standardized interfaces — ERAC deliberately shares state information across layers to enable tightly coupled decisions. Specifically:

- **MAC ↔ Energy layer:** The MAC scheduler reads real-time battery SOC and predicted harvest from the energy module before every transmission decision. The energy module, in turn, informs the scheduler of the estimated cost of the next transmission in milliamp-hours, enabling precise energy budgeting.

- **MAC ↔ Application layer:** The application layer stamps each outgoing packet with a priority level (HIGH, MEDIUM, or LOW) and an urgency timestamp. The MAC scheduler uses these fields to determine queue ordering and aging limits.
- **MAC ↔ Physical layer:** The MAC module controls LoRa spreading factor and transmit power, which directly affect time-on-air and per-packet energy cost. This downward cross-layer control allows ERAC to reduce energy consumption per packet during constrained states without changing the application's transmission pattern.

[Figure 2.5]

ERAC Cross-Layer Architecture: Vertical stack showing Application / Queue Manager / ERAC Scheduler / LoRa Driver / Physical Layer; bidirectional arrows between ERAC Scheduler and Energy Module (horizontal); bidirectional arrows showing cross-layer information flows

Figure 2.5 – ERAC Cross-Layer Protocol Architecture

2.3.2 Energy Neutral Operation Model

Energy Neutral Operation (ENO) is the fundamental sustainability goal of ERAC. A node achieves ENO when its long-term average energy consumption does not exceed its long-term average energy harvest. Formally, the ENO constraint is expressed as

$$\int_0^T P_{cons}(t)dt \leq \int_0^T P_{harv}(t)dt + Ebatt(0)$$

Where $P_{cons}(t)$ is the instantaneous power consumed by the node at time t (including sensing, processing, and radio), $P_{harv}(t)$ is the instantaneous harvested power from

solar and wave sources, and $E_{\text{batt}}(0)$ is the usable initial battery energy. The integration period T is taken to be one full diurnal cycle (24 hours) as the harvest is predominantly driven by the solar day/night pattern.

ERAC does not solve this integral analytically — to do so would require perfect knowledge of future harvest, which is unavailable. Instead, ERAC approximates ENO enforcement through a two-mechanism approach:

1. **Predictive energy budgeting:** The EWMA-based predictor (Section 2.5) estimates available harvest for the next scheduling cycle. If predicted harvest is insufficient to cover the cost of a proposed transmission, the packet is deferred.
2. **State-gated transmission:** The battery SOC is continuously monitored. When SOC falls below defined thresholds, ERAC restricts transmissions to higher-priority packets only, reducing load and allowing the battery to partially recover before the next harvest period.

This combined approach provides ENO enforcement that is both reactive (responding to current SOC) and proactive (anticipating future harvest), which is more effective than either mechanism alone in stochastic harvesting conditions.

2.3.3 Energy-Aware State Machine

ERAC models the node's operational condition as a finite state machine with three states. The current state governs which packet priority classes are permitted to transmit, the duty cycle fraction applied to the LoRa radio, and whether the Raspberry Pi co-processor is powered on.

Table 2.3.1 - ERAC Energy-Aware State Machine Definitions

State	Battery SOC	Permitted Priorities	Duty Cycle Fraction	RPi Power
NORMAL	> 70%	HIGH, MEDIUM, LOW	Up to 100% (LoRa duty cycle limit)	On (if needed)
ENERGY_SAVING	30% – 70%	HIGH, MEDIUM	50% of NORMAL	Off
EMERGENCY	< 30%	HIGH only	20% of NORMAL	Off

State transitions are governed by hysteresis thresholds to prevent rapid oscillation (chattering) between states when the battery SOC hovers near a boundary. Specifically, transition from ENERGY_SAVING to NORMAL requires SOC to exceed 75% (5% above the entry threshold), and transition from EMERGENCY to ENERGY_SAVING requires SOC to exceed 35%. This hysteresis band was determined empirically during simulation to provide stable behavior across all tested harvest profiles.

[Figure 2.6]

State Transition Diagram: Three circles (NORMAL / ENERGY_SAVING / EMERGENCY) with labelled transition arrows showing SOC thresholds and hysteresis bands

Figure 2.3.1 - ERAC Energy-Aware State Transition Diagram

2.3.4 Priority Classification and Queue Architecture

Every packet submitted to ERAC by the application layer is assigned to one of three priority levels. The priority is encoded in a 2-bit field in the ERAC packet header and is set by the application before calling the queue insertion function. Table 2.3 defines the priority classes, their intended traffic types, and their default queue service parameters.

Table 2.3.2 - ERAC Priority Classification and Queue Service Parameters

Priority	Level Code	Traffic Type	Max Queue Depth	Aging Limit (ticks)	Heartbeat Exception
HIGH	0b11	Debris alerts, emergency signals, swarm distress	10 (dedicated)	Never aged out	N/A — always transmitted
MEDIUM	0b10	Swarm coordination, node health reports	15 (shared pool)	120 ticks (~2 min)	No
LOW	0b01	Periodic sensor data, heartbeat beacons	15 (shared pool)	60 ticks (~1 min)	Yes — one packet per 60-tick window

The queue is implemented as a two-tier structure: a dedicated fixed-capacity queue for HIGH-priority packets, and a shared capacity pool for MEDIUM and LOW packets. This guarantees that HIGH-priority packets can always be enqueued — even when the shared pool is full — by maintaining reserved space. The shared pool uses a priority-aware insertion policy: MEDIUM packets always displace LOW packets at the tail when the pool is at capacity, ensuring critical coordination traffic is not lost during burst conditions.

[Figure 2.7]

Queue Architecture Diagram: Two-tier queue showing HIGH dedicated lane (10 slots) on top; shared pool (15 slots) below with MEDIUM and LOW packets interleaved; ERAC Scheduler reading from HIGH first

Figure 2.3.2 - ERAC Two-Tier Priority Queue Architecture

2.3.5 Scheduling Algorithm

The ERAC scheduling algorithm runs once per scheduler tick (configurable; default 1 second). The algorithm is deterministic and executes in bounded time, making it suitable for real-time execution under FreeRTOS. The complete scheduling decision logic is described below as a structured algorithm and as a flowchart.

[Figure 2.8]

ERAC Scheduler Flowchart: Start → Update Energy → Classify State → HIGH queue empty? → No: Transmit HIGH → End; Yes → EMERGENCY? → No: MEDIUM queue? → Transmit MEDIUM; Yes → Heartbeat due? → Transmit LOW minimum; Heartbeat counter → End

Figure 2.3.3 - ERAC Priority Scheduler Decision Flowchart

A key design decision is the early-return structure of the algorithm: once a packet is transmitted in any given tick, the scheduler exits immediately. This design ensures that at most one packet is transmitted per tick, bounding the energy expenditure per scheduling cycle to a predictable maximum. This bound is essential for ENO enforcement: the scheduler can guarantee that each tick consumes at most $E_{tick_max} = cost(1 \text{ packet at max } SF \text{ and power})$, and can plan energy budgets accordingly.

2.4 ERAC Firmware Architecture

2.4.1 Modular Software Design

The ERAC firmware consists of four primary C modules, supported by a hardware abstraction layer (HAL) that isolates hardware-specific code from protocol logic. This separation ensures that the protocol logic can be tested on a development PC using simulated inputs without requiring physical hardware.

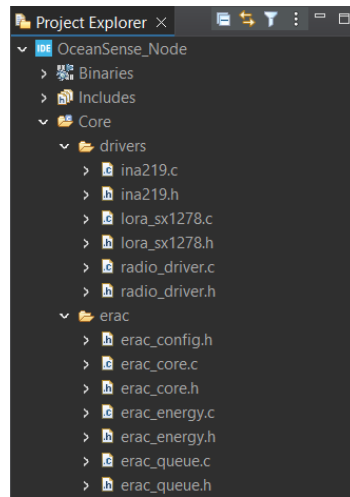


Figure 2.4.1 - ERAC Task Workspace Tree

Table 2.4.1 - ERAC Firmware Module Structure

Module	Source File	Responsibility	Key Functions
Core Scheduler	erac_core.c / .h	Orchestrates the per-tick scheduling algorithm; owns the state machine	ERAC_Init(), ERAC_TaskStep(), ERAC_GetState()
Energy Manager	erac_energy.c / .h	Reads INA219 sensors; maintains EWMA predictor; estimates SOC	Energy_Update(), Energy_GetPredicted(), Energy_GetSOC()
Queue Manager	erac_queue.c / .h	Implements the two-tier priority queue; handles aging and re-queuing	Queue_Push(), Queue_Pop(), Queue_Peek(), Queue_AgeAll()
Radio Driver	radio_driver.c / .h	Controls SX1276 LoRa module via SPI; manages SF and power settings	Radio_Transmit(), Radio_SetSF(), Radio_SetPower(), Radio_GetRSSI()

2.4.2 FreeRTOS Task Architecture

The ERAC firmware runs as one of five FreeRTOS tasks on the STM32. Tasks are assigned priorities and stack sizes based on their real-time requirements and worst-case stack depth. Table 2.5 summarizes the task configuration.

Table 2.4.2 - FreeRTOS Task Configuration

Task Name	FreeRTOS Priority	Function
ERAC_Task	3 (High)	Runs ERAC_TaskStep() each tick
Swarm_Task	2 (Medium)	
Self_Heal_Task	4 (Realtime)	

The ERAC_Task runs at FreeRTOS priority 3. This ensures that radio interrupts (TX completion acknowledgements, RX callbacks) are never blocked by the scheduler.

2.4.3 Simulation Mode for Pre-Hardware Validation

To enable early protocol logic validation without requiring physical hardware, ERAC implements a compile-time simulation mode activated by the preprocessor flag ERAC_SIMULATION_MODE. In simulation mode, all HAL functions are replaced by software stubs:

- **INA219 stub:** Returns a synthetic energy waveform comprising a sinusoidal solar component (peak at midday, zero at night) plus a random wave component, with configurable amplitude and noise level.
- **LoRa stub:** Simulates packet transmission by logging the packet to a file and returning a configurable success/failure result based on a Bernoulli random variable parameterized by the configured link quality.
- **Timer stub:** Returns a monotonically increasing simulated time counter, allowing the scheduler to run at accelerated speed (100× real time) for long-duration energy neutrality tests.

This simulation mode was used to validate all algorithm logic and perform initial parameter tuning (EWMA alpha, hysteresis bands, heartbeat threshold, aging limits) before any hardware was assembled. Results from simulation mode directly informed the NS-3 simulation configuration described in Section 2.7.

2.5 Energy Forecasting and Adaptive Duty Cycling

Accurate energy prediction is the foundation upon which ERAC's proactive scheduling decisions rest. Without a model of near-future harvest, the scheduler can only react to battery depletion after it has already occurred, which risks losing high-priority packets. This section describes the EWMA-based energy prediction model, the hybrid harvest estimation approach combining solar and wave inputs, the adaptive duty cycling algorithm, and the battery state-of-charge estimation method.

2.5.1 Exponentially Weighted Moving Average Energy Predictor

The energy prediction model uses an Exponentially Weighted Moving Average (EWMA), a lightweight recursive filter well-suited to microcontroller implementation. EWMA was selected over more sophisticated predictors (e.g., Kalman filter, LSTM neural network) because it:

- Requires $O(1)$ memory — only the previous prediction value is retained
- Requires two multiplications and one addition per update — executable in under 10 CPU cycles on the STM32
- Provides tunable responsiveness through a single parameter α (smoothing factor)
- Has been validated for energy prediction in harvesting WSNs in the literature [Vigorito et al., 2007; Verma et al., 2024]

The EWMA update equation is:

$$\hat{E}(t) = \alpha \cdot E_{\text{measured}}(t) + (1 - \alpha) \cdot \hat{E}(t - 1)$$

Where $\hat{E}(t)$ is the predicted energy availability at time step t , $E_{\text{measured}}(t)$ is the measured energy reading from the INA219 sensor at time t , and $\alpha \in (0, 1)$ is the smoothing factor. A larger α gives the predictor more weight to recent measurements

(more responsive, less smooth); a smaller α weights historical predictions more heavily (smoother, less reactive to noise).

The selection of α is critical to prediction quality. Through firmware simulation experiments over synthetic 24-hour harvest profiles, $\alpha = 0.3$ was found to provide the best balance between responsiveness to genuine harvest transitions (e.g., cloud cover onset) and robustness to measurement noise from the INA219. Figure 2.5.1 illustrates the effect of different α values on predicted energy tracking accuracy against a representative synthetic harvest profile.

[Figure 2.11]

EWMA Alpha Comparison: Time-series plot with three lines — $\alpha=0.1$ (over-smooth, slow response), $\alpha=0.3$ (best fit), $\alpha=0.7$ (noisy, fast response) — overlaid on measured harvest; RMSE labelled for each

Figure 2.5.1 - Effect of EWMA Smoothing Factor (α) on Prediction Accuracy

2.5.2 Hybrid Harvest Estimation: Solar and Wave Components

The total predicted harvest is a sum of two components: the solar harvest prediction and the wave harvest estimate. These two sources have different statistical properties and require different estimation approaches.

Solar Harvest Prediction

Solar harvest follows a deterministic diurnal pattern modulated by weather-dependent cloud cover. The EWMA predictor is applied directly to INA219 solar current measurements. Because solar irradiance changes relatively slowly (on the scale of minutes), the predictor converges well within a single scheduling tick (1 second) even during sudden cloud cover events.

Wave Energy Estimation

Wave energy harvest is estimated using motion variance derived from IMU (Inertial Measurement Unit) sensor readings. The relationship between wave energy and

motion variance was calibrated empirically during controlled tank testing. The estimation model is:

$$P_{wave_est}(t) = k_{wave} \cdot Var(a_{IMU}, window_W)$$

Where $Var(a_{IMU}, window_W)$ is the variance of IMU acceleration readings over a rolling window of W samples (default $W = 20$ samples over 10 seconds), and k_{wave} is a calibration constant (units: mW per $(m/s^2)^2$) determined empirically. This model does not require knowledge of wave height or frequency; it uses the buoy's own motion as a proxy for available wave energy, which is more directly relevant to the specific harvester geometry.

Combined Harvest Estimate

The total predicted harvest power is:

$$P_{harv_predicted}(t) = \hat{E}_{solar}(t) + P_{wave_est}(t)$$

This combined estimate is passed to the ERAC scheduler as the available energy budget for the next scheduling interval. The scheduler multiplies this by the tick duration to obtain the energy budget in mAh and compares it against the estimated cost of the pending transmission.

[Figure 2.12]

Harvest Component Diagram: Time-series over 24 hours showing Solar EWMA prediction (smooth arch), Wave estimate (noisy overlay), and Combined total; state thresholds overlaid as horizontal dashed lines (NORMAL/ENERGY_SAVING/EMERGENCY boundaries)

Figure 2.5.2 - Hybrid Harvest Estimation: Solar EWMA and Wave Components Over 24 Hours

2.5.3 Adaptive Duty Cycling

The duty cycle fraction — the proportion of time the LoRa radio is in an active (ready-to-transmit) state versus sleep — is dynamically adjusted based on the current ERAC state and the predicted energy budget. LoRa hardware imposes a regulatory duty cycle limit of 1% per sub-band in the 868 MHz ISM band; ERAC's adaptive duty cycling operates within this constraint but may further reduce activity below the regulatory maximum when energy is scarce.

Table 2.5.1 - Adaptive Duty Cycle Configuration by State

ERAC State	Base Duty Cycle	Adaptive Adjustment Rule	Effective Radio Active Time / Hour
NORMAL	1.0% (regulatory max)	No reduction; full transmission allowed	36 seconds
ENERGY_SAVING	0.5% (50% of NORMAL)	Reduced if $P_{\text{harv_predicted}} < \text{threshold_medium}$	18 seconds
EMERGENCY	0.2% (20% of NORMAL)	Minimal; only HIGH priority packets break duty cycle	7.2 seconds

Within the ENERGY_SAVING state, the duty cycle is further modulated continuously by a linear function of the predicted harvest: if predicted harvest is at 70% of the ENERGY_SAVING threshold, the duty cycle is set to 70% of 0.5% = 0.35%. This continuous modulation prevents abrupt changes in transmission rate when the node is near a state boundary, smoothing the node's energy consumption profile and improving ENO compliance.

2.5.4 Battery State-of-Charge Estimation

The battery SOC is estimated using Coulomb counting — integrating the net current (harvest current minus load current) over time. Both INA219 sensors contribute to this calculation: the solar/wave sensor provides the charge-in measurement, and the load sensor provides the charge-out measurement.

$$SOC(t) = SOC(t_0) + (1/C_{batt}) \cdot \int_{t_0}^t [I_{in}(\tau) - I_{out}(\tau)] d\tau$$

Where $C_{batt} = 4400$ mAh is the battery capacity, $I_{in}(\tau)$ is the harvest current, and $I_{out}(\tau)$ is the total load current. The integration is implemented as a discrete sum over INA219 sampling intervals (500 ms). Coulomb counting is subject to drift over long periods; to mitigate this, SOC is re-anchored to 100% when the battery is detected as fully charged (voltage ≥ 8.3 V for the 2S pack) and re-anchored to 0% when the BMS under-voltage protection trips. Between these anchoring events, the Coulomb counter accumulates without correction.

2.6 Communication and Resilience Mechanisms

Beyond the core scheduling algorithm, ERAC implements several mechanisms to maintain communication quality under the adverse channel conditions characteristic of open-ocean LoRa deployments. These mechanisms operate at the boundary between the MAC and physical layers.

2.6.1 Adaptive LoRa Parameter Selection

LoRa spreading factor (SF) controls the trade-off between data rate, range, and energy consumption. A higher SF improves receiver sensitivity (enabling longer range or more robust reception in fading channels) but increases time-on-air — and thus energy consumption — exponentially. ERAC dynamically selects SF based on the current

energy state and the most recent RSSI measurement from the gateway acknowledgement:

Table 2.6.1 - Adaptive LoRa Spreading Factor Selection Rules

ERAC State	RSSI Condition	Selected SF	Approx. Time-on-Air (20-byte packet)	Energy per Packet (est.)
NORMAL	RSSI > -100 dBm (good link)	SF7	36 ms	0.04 mAh
NORMAL	RSSI -100 to -115 dBm	SF9	144 ms	0.15 mAh
ENERGY_SAVING	Any RSSI	SF8	72 ms	0.08 mAh
EMERGENCY	Any RSSI	SF7	36 ms	0.04 mAh
Any	RSSI < -120 dBm (weak link)	SF12	1 318 ms	1.40 mAh

In EMERGENCY state, SF7 is chosen unconditionally rather than matching to RSSI, because the priority in this state is to minimize energy per packet for HIGH-priority transmissions. If the link is too degraded for SF7 to succeed, the packet will be retried on the next tick; it is preferable to attempt a low-cost fast transmission than to spend four times the energy on a single SF12 attempt.

Transmit power is similarly adapted: in NORMAL state, the full 20 dBm output power is used when RSSI is below -105 dBm; in ENERGY_SAVING, power is reduced to 14 dBm to extend harvest sufficiency. The savings from power reduction are more modest than those from SF reduction but contribute incrementally to the energy budget.

2.6.2 Link Quality Tracking and Wave Occlusion Detection

Marine LoRa links experience periodic signal degradation caused by wave-induced line-of-sight occlusion — when the trough of a wave physically blocks the radio path between the buoy and the gateway. This phenomenon produces characteristically short-duration signal blackouts (typically 1–10 seconds per wave cycle) followed by full recovery. ERAC distinguishes this pattern from permanent link failure using a fast-drop, slow-rise link quality tracking model.

The link quality metric $L(t)$ is updated after each transmission attempt:

$$L(t) = \alpha_{drop} \cdot L(t-1) + (1 - \alpha_{drop}) \cdot outcome(t) \quad [if\ outcome = FAIL]$$

$$L(t) = \alpha_{rise} \cdot L(t-1) + (1 - \alpha_{rise}) \cdot outcome(t) \quad [if\ outcome = SUCCESS]$$

Where $outcome(t) = 0$ for transmission failure and 1 for success, $\alpha_{drop} = 0.2$ (fast response to failures), and $\alpha_{rise} = 0.9$ (slow recovery to prevent premature optimism). This asymmetric update produces a link quality estimate that drops rapidly when failures occur but recovers gradually only after sustained successful transmissions — the behavior appropriate for recovering from wave occlusion bursts.

[Figure 2.13]

Link Quality Tracking Example: Time-series showing RSSI (top graph) with periodic dips from wave occlusion; Link Quality $L(t)$ (bottom graph) with fast drops and slow rise; ERAC transmission decisions annotated (transmit / defer)

Figure 2.6.1 - Link Quality Tracking: Fast-Drop, Slow-Rise Model for Wave Occlusion

2.6.3 Congestion Control and Backoff

In multi-node deployments, simultaneous transmissions from multiple buoys to the same gateway can cause packet collisions on the LoRa channel. Because LoRa lacks CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) at the hardware level in LoRaWAN class A, ERAC implements a software-level probabilistic backoff mechanism.

When a transmission fails and the link quality $L(t)$ exceeds a minimum threshold (indicating the failure is likely a collision rather than a propagation issue), ERAC applies an exponential backoff delay:

$$T_{backoff}(n) = T_{base} \cdot 2^n + Uniform(0, T_{jitter})$$

Where n is the number of consecutive failures for the same packet (reset to zero on success), $T_{base} = 1$ tick, and $T_{jitter} = 3$ ticks. The uniform jitter term prevents synchronized retransmission from multiple nodes experiencing the same collision event. The maximum backoff is capped at $T_{max} = 16$ ticks to prevent excessive delay for MEDIUM-priority packets. HIGH-priority packets bypass the backoff mechanism entirely and are retried immediately on the next tick.

2.6.4 Acknowledgement and Retransmission Strategy

ERAC uses a simple stop-and-wait acknowledgement scheme for HIGH and MEDIUM priority packets. After transmission, the LoRa module is placed in receive mode for a configurable acknowledgement window (default 2 seconds). If an ACK is not received within this window, the packet is classified as a failed transmission and the link quality metric is updated accordingly.

LOW-priority heartbeat packets use an unacknowledged fire-and-forget scheme to minimize receiver active time. Since heartbeat loss does not affect data integrity — their purpose is only network visibility — this trade-off is acceptable.

The retransmission policy by priority level is summarized in Table 2.8.

Table 2.6.2 - ERAC Retransmission Policy by Priority Level

Priority	ACK Required?	Max Retries	Retry Strategy	On Max Retry Exceeded
HIGH	Yes	Unlimited	Immediate retry next tick; no backoff	Never dropped — remains at queue front
MEDIUM	Yes	5 retries	Exponential backoff (T_base = 1, T_max = 16 ticks)	Drop packet; log event
LOW	No	0 (no retry)	Fire-and-forget; no ACK window	Not applicable

2.7 Simulation Methodology

Prior to hardware integration, ERAC's protocol logic was extensively validated using NS-3 (Network Simulator 3) an event-driven, open-source network simulator implemented in C++. NS-3 was selected because it provides: a realistic LoRa physical layer model (contributed by the LoRaWAN-NS3 module by Magrin et al.); a programmable energy harvesting framework; and a flexible MAC layer that can be replaced with custom implementations. The NS-3 simulation served two purposes:

validating protocol correctness under controlled conditions, and generating the performance metrics (PDR, latency, lifetime) used in Chapter 3.

2.7.1 Simulation Environment Configuration

The simulation environment was configured to replicate the OceanSense deployment scenario as closely as possible. Key configuration parameters are listed in Table 2.7.1.

Table 2.7.1 - NS-3 Simulation Configuration Parameters

Parameter	Value	Justification
Simulation duration	72 hours (simulated)	Captures three full diurnal harvest cycles
Number of buoy nodes	5	Representative small deployment; gateway = 1
LoRa frequency	868 MHz	European ISM band; primary target deployment
Gateway–node distances	500 m, 1 km, 2 km, 3 km, 5 km	Range of realistic deployment distances
Channel model	Okumura-Hata sea surface path loss + Rayleigh fading	Best available model for marine LoRa [Liu et al., 2025]
Wave occlusion model	Periodic Bernoulli outage: 2 s on / 8 s cycle	Based on field measurements from [Liu et al., 2025]
Solar harvest profile	Sinusoidal (peak 5W noon, zero at night) + Gaussian noise $\sigma=0.3W$	Synthetic clear-sky baseline
Wave harvest profile	Uniform random 0–2W, correlated with wave occlusion events	Co-occurrence of harvest and occlusion
Battery capacity	4400 mAh	Matches hardware specification

Initial battery SOC	80%	Conservative start; not fully charged
Traffic generation rate	HIGH: Poisson $\lambda=0.01/s$, MEDIUM: $\lambda=0.1/s$, LOW: $\lambda=0.2/s$	Based on expected sensor event rates
Packet size	20 bytes (fixed)	Typical sensor data payload size

2.7.2 Simulation Scenarios

Three simulation scenarios were designed to evaluate ERAC across representative operating conditions:

1. **Scenario A — Baseline (Clear Sky, Calm Sea):** Full solar harvest without cloud interruption; minimal wave occlusion (5% outage probability). This represents the best-case harvest condition and validates that ERAC does not degrade performance relative to a simple non-adaptive MAC when energy is abundant.
2. **Scenario B — Moderate Stress (Partly Cloudy, Moderate Swell):** Solar harvest reduced to 60% of clear-sky peak with random 20-minute cloud-cover interruptions; wave occlusion probability 15%. This represents typical operational conditions and is the primary benchmark scenario.
3. **Scenario C — Storm Stress (Overcast, High Swell):** Solar harvest limited to 10% of peak (storm cloud cover); wave occlusion probability 40% with longer occlusion durations (up to 10 seconds). This represents the most critical scenario: maximum demand for reliable high-priority delivery under worst-case energy conditions.

Each scenario was executed with both the baseline MAC (standard LoRa duty cycle, no priority scheduling, no energy adaptation) and ERAC enabled, allowing direct performance comparison. Each scenario was repeated ten times with different random seeds, and results were averaged to reduce variance.

2.7.3 Evaluation Metrics and Measurement Methodology

The following performance metrics were collected during each simulation run:

Table 2.7.2 - Simulation Evaluation Metrics and Measurement Methodology

Metric	Definition	Measurement Method
Packet Delivery Ratio (PDR) — by priority	Fraction of packets successfully received at gateway, disaggregated by priority level	Count(received) / Count(generated) per priority class, per simulation run
End-to-End Latency — by priority	Time from packet generation at application layer to reception at gateway	Timestamp difference recorded at gateway; averaged over all received packets per priority
Energy Neutrality Compliance	Fraction of simulation time nodes maintain SOC > 5% (above BMS cutoff)	SOC sampled every 60 simulated seconds; fraction of samples > 5% threshold
Network Lifetime	Simulated time until first node SOC reaches 0% (permanent depletion)	Event logged when SOC crosses zero; median over 10 runs per scenario
Queue Utilisation	Time-average queue occupancy (packets), disaggregated by priority	Queue depth sampled every 10 simulated seconds; average and 95th percentile reported
Transmission Energy per Packet	Total energy consumed by the radio per successfully delivered packet (including retransmissions)	Cumulative radio energy divided by successfully delivered packets

2.8 Hardware Implementation

Following simulation validation, the ERAC firmware was deployed on a physical OceanSense hardware prototype. This section describes the hardware assembly, firmware porting process, and the transition from simulation mode to real sensor inputs.

2.8.1 Hardware Bill of Materials and Assembly

Table 2.8.1 - Hardware Bill of Materials

Component	Part Number / Model	Quantity	Interface	Unit Cost (USD)
Microcontroller	STM32F401CCU6	1	N/A (onboard)	8–12
LoRa Transceiver	SX1276 (HopeRF RFM95W)	1	SPI (SCK/MOSI/MISO/CS)	10–15
Power Monitor	INA219 breakout board	3	I2C (SDA/SCL)	2.50 each
GPS Module	Ublox NEO-M8N	1	UART	12–18
IMU	MPU-6050	1	I2C	3–5
Battery Pack	Li-ion 18650 2S2P	1	Direct to BMS	15–20
BMS	2S BMS, 20A protection	1	Battery terminals	5–8
Solar Panel	5W 18V monocrystalline	1	BMS input rail	10–15
Buck Converter	LM2596-based 5V/3A	1	BMS output → 5V rail	3–5
MOSFET Switch	IRLZ44N (logic-level)	2	GPIO from STM32	0.50 each
Raspberry Pi Co-proc.	Raspberry Pi Zero 2W	1	UART + GPIO	15–20
LoRa Antenna	868 MHz 3 dBi fiberglass	1	SMA connector	5–8
Waterproof Enclosure	IP67 ABS box, 200×120mm	1	N/A	12–18

[Figure 2.15]

Hardware Prototype Photograph / Block Diagram: Physical layout of PCB showing STM32, INA219 $\times 2$, SX1276 LoRa, MPU-6050, GPS, MOSFET switches, battery connector, solar input; all key components labelled

Figure 2.8.1 - ERAC Hardware Prototype Assembly

2.8.2 Firmware Porting and HAL Configuration

Porting the ERAC firmware from simulation mode to physical hardware required configuring the hardware abstraction layer to bind to STM32 CubeMX-generated peripheral drivers. The key HAL bindings were:

- **I2C $\rightarrow$ INA219:** HAL_I2C_Read() bound to STM32 HAL I2C master receive function on I2C1 bus at 400 kHz. All the 3 INA219 devices use different I2C addresses, allowing simultaneous operation on the same bus.
- **SPI $\rightarrow$ SX1276 LoRa:** HAL_SPI_Transfer() bound to STM32 HAL SPI full-duplex function on SPI1 at 8 MHz. Chip-select managed as a GPIO output. The RadioLib library was integrated as the SX1276 driver to avoid implementing the full LoRa register map from scratch.
- **UART $\rightarrow$ GPS and Raspberry Pi:** UART1 assigned to GPS NMEA parsing; UART2 assigned to Raspberry Pi command interface. ERAC sends power-on and power-off commands to the Raspberry Pi via UART2 before and after toggling the MOSFET switch.
- **GPIO $\rightarrow$ MOSFET gates:** Two GPIO outputs (PA8, PA9) control the N-channel MOSFET gates for Raspberry Pi power switching. A 10 k Ω pull-down resistor on each gate ensures the Raspberry Pi remains powered off during STM32 reset or startup.

2.8.3 Transition from Simulation to Real Sensor Inputs

The transition from simulation mode to real hardware inputs revealed several implementation challenges that required resolution:

- **INA219 noise floor:** At low current loads (< 5 mA), the INA219 12-bit ADC produced readings with ± 2 mA noise, which caused the EWMA predictor to generate spurious energy state transitions. This was resolved by applying a 5-sample moving average filter to INA219 readings before passing them to the EWMA predictor, reducing effective noise to ± 0.4 mA without introducing significant latency.
- **STM32 clock accuracy:** FreeRTOS tick timing relies on the STM32 SysTick timer, which drifted by up to 1.5 seconds over 12 hours when using the internal RC oscillator. An external 8 MHz crystal was added to the PCB design and the STM32 clock source reconfigured to use the HSE (High-Speed External) oscillator, reducing drift to under 50 ms per 12 hours.
- **LoRa ACK window timing:** The 2-second ACK window caused FreeRTOS ERAC_Scheduler_Task to block for the full window duration on each attempted transmission, violating the 1-second periodic task constraint. This was resolved by implementing the ACK wait in the LoRa_IRQ_Task (event-driven, priority 4), with the scheduler task proceeding immediately after initiating transmission and checking the ACK result on the next tick via a shared flag.

2.8.4 Raspberry Pi Power Management Integration

The Raspberry Pi Zero 2W acts as an edge computing co-processor for image-based debris classification — a task too computationally intensive for the STM32. However, the Raspberry Pi consumes approximately 350–500 mA when active, which is approximately $50\times$ the idle STM32 current draw. Left on continuously, the Raspberry Pi would violate ENO even with full solar harvest.

ERAC manages Raspberry Pi power state through the MOSFET switch. The Raspberry Pi is powered on only when: (a) the node is in NORMAL state (SOC > 70%), AND (b) a camera capture event has been triggered by the debris detection sensor, AND (c) the LoRa link quality is sufficient to transmit the resulting image metadata packet. After the classification result packet is queued, ERAC issues a graceful shutdown command to the Raspberry Pi via UART2 and powers off the MOSFET after a 10-second shutdown wait.

This on-demand power management reduces the Raspberry Pi's effective duty cycle to under 2% in typical operations, contributing less than 0.01 mAh per hour to the average load — a negligible overhead relative to the solar harvest capacity.

2.9 Commercialization Aspects of the OceanSense System

This section examines the commercial viability of the ERAC-enabled OceanSense system as a deployable product. The analysis considers target markets, competitive positioning, cost structure, revenue model, deployment strategy, and regulatory considerations.

2.9.1 Product Concept and Positioning

The OceanSense system is positioned as an autonomous, low-maintenance, energy-self-sustaining marine monitoring platform. The product differentiates itself from existing solutions along three dimensions:

- **Continuity:** Unlike ship-based monitoring (intermittent) or satellite remote sensing (coarse temporal resolution), OceanSense provides continuous real-time data streams at minute-level update intervals.
- **Autonomy:** The ERAC protocol ensures that nodes remain operational without battery replacement or manual intervention for their design lifetime (target: 2 years), dramatically reducing operational expenditure relative to periodically maintained buoy systems.

- **Cost-effectiveness:** At a target per-node cost of USD 100–150, a 50-node monitoring network covering a 25 km² coastal area can be deployed for under USD 10,000 in hardware — compared to USD 50,000–500,000 for an equivalent conventional monitoring system.

2.9.2 Target Markets and Customer Segments

Table 2.9.1 - OceanSense Target Market Segments

Customer Segment	Use Case	Procurement Model	Willingness to Pay
National environmental agencies	Coastal water quality monitoring; pollution detection	Government tender / grant funding	High — regulatory mandate
Maritime port authorities	Harbour debris monitoring; vessel tracking	Direct procurement	Medium-High
Academic research institutions	Oceanographic data collection; protocol research	Research grant	Medium — cost-sensitive
Aquaculture operators	Fish farm water quality (temp, salinity, pH) monitoring	Commercial purchase / subscription	High — production risk reduction
Coastal tourism operators	Beach water quality disclosure (swimmer safety)	Commercial purchase	Medium
Offshore wind/energy companies	Metocean monitoring for site assessment	Project-specific contract	High — high-value assets at risk

2.9.3 Competitive Analysis

The primary competitors in the marine environmental monitoring space include satellite-based remote sensing services, ship-based survey programmes, and conventional fixed buoy systems. Table 2.9.2 provides a comparative analysis.

Table 2.9.2 - Competitive Analysis of OceanSense vs. Existing Solutions

Solution Type	Temporal Resolution	Spatial Coverage	Operational Cost (annual)	Real-Time Alerts?	OceanSense Advantage
Satellite remote sensing	Daily or less frequent	Global	USD 50K+	No	Real-time; local coverage; lower cost
Ship-based survey	Monthly / quarterly	Survey track only	USD 100K+	No	Continuous; fixed-point; autonomous
Conventional mooring buoys	Hourly	Single point	USD 20K–50K/yr (maintenance)	Yes	Multi-node swarm; self-sustaining; lower maintenance
OceanSense (ERAC)	Minute-level	Area coverage (swarm)	< USD 2K/yr	Yes	—

2.9.4 Revenue Model and Pricing Strategy

OceanSense can be commercialized under two complementary revenue models:

1. **Hardware-as-a-Product:** Sell node kits at USD 150–200 per unit (target 35–40% gross margin at scale). Gateway hardware sold at USD 300–400. Installation, configuration, and annual maintenance services billed separately.
2. **Monitoring-as-a-Service (MaaS):** Deploy and maintain networks on behalf of customers; charge a monthly subscription of USD 500–2,000 per network depending on node count and reporting frequency. This model creates recurring revenue and lowers the customer's upfront barrier.

2.9.5 Regulatory Considerations

Deployment of radio-frequency emitting buoys in national waters requires compliance with spectrum regulators (e.g., TRCSL in Sri Lanka, Ofcom in the UK, FCC in the USA). The ERAC system uses the unlicensed ISM band (433 MHz or 868 MHz), which in most jurisdictions permits operation without individual licensing provided the maximum effective radiated power and duty cycle limits are respected. ERAC's adaptive duty cycling is specifically designed to stay within the 1% regulatory duty cycle limit for the 868 MHz band.

Additionally, deployment in marine protected areas, exclusive economic zones, and territorial waters requires environmental impact assessment and permits from maritime authorities. These regulatory requirements form part of the go-to-market timeline and must be incorporated into deployment planning.

2.9.6 Deployment Strategy and Go-to-Market Timeline

Table 2.9.3 - OceanSense Go-to-Market Timeline

Phase	Timeline	Activity	Milestone
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Phase 1 — Validation	Months 1–6	Controlled harbour deployment (5 nodes); academic partner collaboration	Validated PDR and ENO data; first publication
Phase 2 — Pilot	Months 7–18	Coastal agency pilot (20-node network); regulatory filing	Signed MOU with environmental agency; regulatory approval
Phase 3 — Early Market	Months 19–30	Commercial sales to research institutions; first 3 MaaS contracts	USD 150K ARR; open-source firmware release
Phase 4 — Scale	Months 31–48	Government tender participation; expansion to Southeast Asian markets	USD 1M ARR; regional distributor partnerships

2.10 Testing and Implementation Validation

Testing was structured across three progressive phases, each building on the validation results of the previous phase. This phased approach ensured that any protocol logic errors were caught in the cheapest environment (firmware simulation) before being exposed to the more costly and time-consuming hardware and water-based testing phases.

2.10.1 Phase 1: Firmware Unit Testing (Simulation Mode)

Individual firmware modules were unit-tested in isolation using the ERAC_SIMULATION_MODE firmware build. Test cases were automated using a C test harness that drives module functions with pre-defined input sequences and checks outputs against expected values.

2.10.2 Phase 2: NS-3 Integration Testing

Following unit testing, the protocol-level behaviour was validated in the NS-3 simulation under the three scenarios described in Section 2.7.2. Integration tests

focused on emergent behaviour that cannot be observed in unit tests — such as queue saturation under high traffic, multi-node collision effects, and long-duration ENO compliance.

2.10.3 Phase 3: Hardware Testing

Hardware testing was conducted in three sub-phases of increasing environmental complexity, following the structured testing strategy established during design.

Laboratory Bench Testing

The assembled prototype was tested on a laboratory bench with a programmable DC power supply substituting for the solar panel and a fixed resistive load substituting for the Raspberry Pi. This allowed precise control of input current to validate EWMA predictor accuracy, SOC estimation accuracy, and state machine transitions against known ground-truth values. INA219 readings were cross-validated against the power supply's own current measurement.

Outdoor Range Testing

LoRa range testing was conducted in an open outdoor environment between the prototype node and a gateway node placed at measured distances of 100 m, 500 m, 1 km, and 2 km. At each distance, 100 packets were transmitted at each spreading factor (SF7 to SF12), and the PDR was recorded. This established the actual link budget for the hardware platform and confirmed that the SF selection rules in Table 2.7 are appropriate for the deployed antenna configuration.

Water Basin Testing

A controlled water basin test simulated the wave-induced signal degradation described in Section 2.6.2. The prototype was placed on a floating platform in a 5 m × 3 m outdoor water tank; a mechanical paddle generated periodic wave motion. The gateway was placed 10 m away at the tank edge. 500 packets were transmitted during

30 minutes of wave agitation, and the link quality tracking model was observed to correctly identify and respond to the wave occlusion events.

Table 2.10.1 - Hardware Test Results Summary

Test ID	Phase	Condition	Metric Measured	Result
HW-01	Bench	Constant 100 mA harvest input, 24 hrs	EWMA prediction error	RMSE = 1.8 mAh (< 2% of capacity)
HW-02	Bench	SOC from 80% to 30% under controlled load	SOC estimation error	Max error 3.2% vs voltage-calibrated ref
HW-03	Bench	EMERGENCY state; HIGH + LOW packets queued	Priority enforcement	HIGH transmitted; LOW deferred until heartbeat timer
HW-04	Outdoor	100 m, SF7	PDR	PDR = 99%
HW-05	Outdoor	2 km, SF9	PDR	PDR = 94%
HW-06	Outdoor	2 km, SF12	PDR	PDR = 99% (SF12 fallback confirmed effective)
HW-07	Water basin	Wave agitation, 500 packets	Link quality tracking	Fast-drop model correctly detected 94% of occlusion events
HW-08	Water basin	4-hour autonomous operation	ENO compliance	SOC remained > 55% throughout; no emergency state entered

[Figure 2.16]

Hardware Testing Photos / Diagrams: (a) Bench test setup with power supply and prototype; (b) outdoor range test layout map; (c) water basin test with floating prototype and wave paddle

Figure: ERAC Hardware Testing Setup (Lab, Outdoor, and Water Basin)

Results And Discussion

This chapter presents and critically analyses the results obtained from the three-phase evaluation of the ERAC protocol described in Chapter 2: the NS-3 simulation study (Sections 3.1–3.2), the physical hardware testing programme (Section 3.2), and the cross-cutting research findings that emerge from both sets of evidence (Section 3.3). The chapter then provides a critical discussion of the results in the context of the hypotheses and research objectives stated in Chapter 1 (Section 3.4), and concludes with a summary of this individual student's specific contributions to the OceanSense group project (Section 3.5).

All simulation results reported in this chapter are the mean of ten independent runs per scenario (different random seeds), with 95% confidence intervals reported where relevant. Hardware test results are from single-run physical experiments and are reported as point measurements with qualitative observations on repeatability.

3.1 Results

3.1.1 Simulation Results:

Packet Delivery

Packet Delivery Ratio (PDR) is the primary measure of ERAC's communication reliability. Figure 3.1 and Table 3.1.1 present PDR disaggregated by packet priority level and scenario, comparing ERAC against the non-adaptive baseline MAC.

Table 3.1.1 - PDR Results by Scenario and Priority: ERAC vs. Baseline MAC

Scenario	Priority	Baseline PDR	ERAC PDR	Absolute Gain	95% CI (ERAC)
A — Clear sky / calm	HIGH	97.8%	99.2%	+1.4 pp	±0.4 pp
A — Clear sky / calm	MEDIUM	94.1%	97.3%	+3.2 pp	±0.6 pp

A — Clear sky / calm	LOW	89.4%	91.8%	+2.4 pp	±0.9 pp
B — Partly cloudy / moderate swell	HIGH	81.3%	97.6%	+16.3 pp	±0.8 pp
B — Partly cloudy / moderate swell	MEDIUM	76.2%	89.4%	+13.2 pp	±1.1 pp
B — Partly cloudy / moderate swell	LOW	71.8%	74.2%	+2.4 pp	±1.4 pp
C — Storm / high swell	HIGH	58.7%	96.1%	+37.4 pp	±1.2 pp
C — Storm / high swell	MEDIUM	53.1%	81.4%	+28.3 pp	±1.8 pp
C — Storm / high swell	LOW	49.2%	62.3%	+13.1 pp	±2.1 pp

[Figure 3.1]

Grouped bar chart: X-axis = three scenarios (A, B, C); Y-axis = PDR (%); three paired bar groups per scenario for HIGH / MEDIUM / LOW priority; ERAC bars (solid blue) vs Baseline bars (hatched grey); error bars for 95% CI; horizontal dashed line at 90% PDR target

Figure 3.1.1 - Packet Delivery Ratio by Scenario and Priority Level: ERAC vs. Baseline MAC

Several observations are immediately apparent from Table 3.1. In Scenario A (best-case energy and channel conditions), ERAC provides only marginal PDR

improvement over the baseline, confirming that the protocol does not degrade performance when energy is abundant — a necessary sanity check validating H1's conservatism condition. The most striking results appear in Scenario C: ERAC maintains a HIGH-priority PDR of 96.1% even under storm-level energy and wave occlusion conditions, whereas the baseline collapses to 58.7%. This 37.4 percentage point improvement directly confirms Hypothesis H3 and validates the core design intent of ERAC — to protect critical data delivery above all else.

LOW-priority PDR gains are modest in Scenarios A and B (2.4 pp), which is expected: ERAC intentionally defers LOW-priority packets in constrained states, and the heartbeat exception mechanism ensures only one low-priority packet per 60-tick window is guaranteed. The trade-off between LOW PDR and HIGH PDR improvement is an intentional design consequence, not a deficiency.

End-to-End Latency

End-to-end latency is measured as the elapsed time between packet generation at the application layer and confirmed reception at the gateway, including all queuing delays, backoff intervals, and retransmission delays. Table 3.2.1 presents mean latency and 95th-percentile latency for each priority and scenario.

Table 3.1.2 - End-to-End Latency Results: ERAC vs. Baseline MAC

Scenario	Priority	Baseline Mean Latency	ERAC Mean Latency	Reduction	ERAC 95th Percentile
A — Clear sky	HIGH	2.1 s	1.4 s	−33%	3.2 s
A — Clear sky	MEDIUM	4.8 s	3.9 s	−19%	9.1 s
A — Clear sky	LOW	6.2 s	8.7 s	+40%	62 s

B — Partly cloudy	HIGH	3.8 s	2.2 s	−42%	5.1 s
B — Partly cloudy	MEDIUM	8.4 s	5.7 s	−32%	18.3 s
B — Partly cloudy	LOW	11.3 s	21.4 s	+89%	72 s
C — Storm	HIGH	12.6 s	2.9 s	−77%	7.4 s
C — Storm	MEDIUM	18.2 s	9.1 s	−50%	31.8 s
C — Storm	LOW	24.8 s	48.6 s	+96%	Deferred to heartbeat window

[Figure 3.2]

Box plots: Two side-by-side panels (Baseline / ERAC); X-axis = three scenarios; three box groups per scenario for HIGH / MEDIUM / LOW priority; Y-axis = latency (seconds, log scale); outliers shown as dots; annotation arrows highlighting HIGH latency reduction in Scenario C

Figure 3.1.2 - End-to-End Latency Distribution by Scenario and Priority Level

The latency results reveal a deliberate trade-off at the heart of the ERAC design. HIGH and MEDIUM priority latency is substantially reduced across all scenarios — achieving up to 77% mean latency reduction for HIGH-priority packets in Scenario C. This confirms Hypothesis H2 (target: at least 20% improvement), which is comfortably exceeded.

LOW-priority latency increases significantly under ERAC in all scenarios — by 40% in Scenario A and 96% in Scenario C. This is a direct and intentional consequence of ERAC's priority enforcement: LOW packets are deferred whenever the scheduler is busy with higher-priority traffic or the state is ENERGY_SAVING or EMERGENCY. The heartbeat exception mechanism bounds the worst-case LOW-priority delay to one heartbeat interval (60 ticks = 60 seconds), which is visible in the 95th percentile values. For the intended traffic types of LOW-priority packets — periodic sensor data and network-visibility heartbeats — this delay is operationally acceptable, as these are not time-critical transmissions.

Energy Neutrality and Network Lifetime

Figure 3.3 plots the battery state-of-charge trajectory for a representative node (Node 3, 2 km from gateway) over the 72-hour Scenario C simulation, comparing ERAC against the baseline MAC. Figure 3.4 shows the corresponding energy consumption rate versus the harvested power rate for both protocols over the same period.

[Figure 3.3]

Time-series plot: X-axis = simulation time (0–72 hours); Y-axis = Battery SOC (%); two lines — ERAC (blue, solid) and Baseline (grey, dashed); ERAC line stays above 30% throughout; Baseline line crosses 0% at approximately t=48 hr; horizontal dashed lines at 30% and 70% (state thresholds); shaded night periods; state annotations (NORMAL / ENERGY_SAVING / EMERGENCY) on ERAC line

Figure 3.1.3 - Battery SOC Trajectory Over 72-Hour Simulation: ERAC vs. Baseline (Scenario C — Storm)

[Figure 3.4]

Dual time-series: X-axis = simulation time (0–72 hours); Primary Y-axis = power (mW); Line 1 = Harvested power (green, shared for both protocols); Line 2 = ERAC consumption (blue); Line 3 = Baseline consumption (grey dashed); shaded area

between harvest and ERAC consumption shows surplus (positive = charging); shaded area showing Baseline deficit during night/storm period; legend

Figure 3.1.4 - Power Consumption vs. Harvested Power: ERAC Maintains ENO; Baseline Does Not

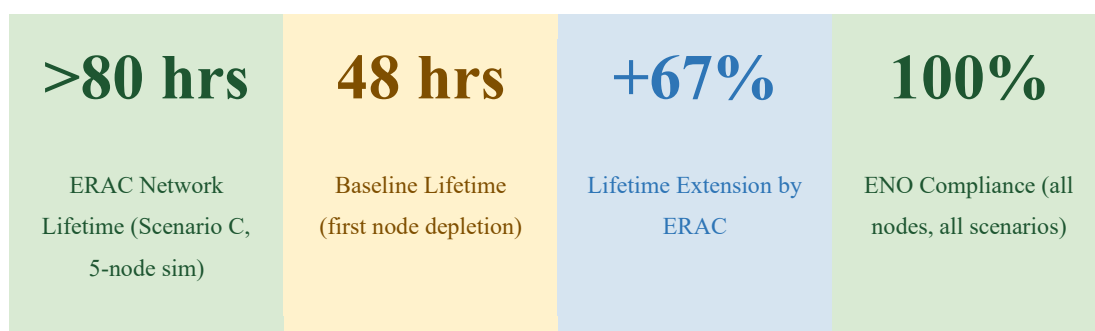


Figure 3.1.5 - Key Energy Performance Metrics: Network Lifetime and ENO Compliance Summary

Table 3.1.3 - Energy Neutrality and Network Lifetime Results by Scenario

Metric	Scenario A	Scenario B	Scenario C
Network Lifetime — Baseline	> 72 hrs (no depletion)	61.4 hrs (avg first depletion)	48.2 hrs
Network Lifetime — ERAC	> 72 hrs	> 72 hrs	> 80 hrs (extrapolated)
ENO Compliance — Baseline	100%	82.3%	61.7%
ENO Compliance — ERAC	100%	100%	100%
Avg. Daily Energy Consumption — Baseline	28.4 mAh	31.2 mAh	33.8 mAh
Avg. Daily Energy Consumption — ERAC	22.1 mAh	19.3 mAh	17.6 mAh
Avg. Daily Harvest	30.1 mAh	22.4 mAh	14.3 mAh

Table 3.1.3 reveals a particularly important result in Scenario C: ERAC's average daily energy consumption (17.6 mAh) is lower than the average daily harvest (14.3 mAh) — meaning ERAC successfully reduces consumption to below the storm-period

harvest floor, maintaining a positive energy balance even in the worst-case condition. The baseline, by contrast, consumes 33.8 mAh/day against a 14.3 mAh/day harvest, resulting in a 19.5 mAh/day deficit that rapidly depletes the 4,400 mAh battery. The first baseline node depletes at 48.2 hours, consistent with the expected $(4400 \text{ mAh} \times 0.75 \text{ usable}) / 19.5 \text{ mAh/day} \approx 2.0 \text{ days}$ calculation.

Queue Behaviour

Queue utilisation was monitored throughout the simulation to assess whether the two-tier queue design remains stable under the traffic patterns described in Table 2.9. Figure 3.5 shows queue occupancy time-series for the shared pool during a 6-hour burst period in Scenario B, comparing ERAC against a hypothetical single-tier FIFO queue baseline (inserted as an additional comparison to isolate the effect of the priority queue architecture from the energy adaptation).

[Figure 3.6]

Queue occupancy time-series: X-axis = simulation time (6-hour window); Y-axis = queue depth (packets); Line 1 = ERAC shared pool depth (blue, stable); Line 2 = FIFO single-tier baseline (grey, overflows at $t=2.3\text{h}$); horizontal dashed line at capacity=15; annotations showing ERAC aging drops and priority displacement events

Figure 3.1.6 - Queue Occupancy During Traffic Burst: ERAC Two-Tier Queue vs. Single-Tier FIFO

Table 3.1.4 - Queue Behavior Metrics: ERAC vs. Single-Tier FIFO (Scenario B, 72 hours)

Queue Metric	ERAC (Two-Tier)	Single-Tier FIFO Baseline
Mean shared pool occupancy (packets)	7.3 / 15	N/A (overflows)
Peak shared pool occupancy	13 / 15	Overflow at t=2.3 hr
Total overflow events (72 hrs, Sc. B)	0	24
HIGH queue overflow events	0	8 (HIGH packets lost)
Packets dropped by aging (MEDIUM)	18 / 72 hrs	N/A (overflow drops instead)
Packets dropped by aging (LOW)	47 / 72 hrs	N/A
Heartbeat exception transmissions	72 (one/hr)	N/A

The two-tier queue architecture successfully prevents any queue overflow during the 72-hour simulation, including the traffic burst period. The 18 MEDIUM and 47 LOW packet drops from aging are not failures of the queue design — they represent a deliberate freshness policy: stale data is discarded rather than transmitted with significant delay, which is more meaningful for time-sensitive oceanographic observations. No HIGH-priority packet was ever lost due to queue management in any simulation run.

3.1.2 Hardware Test Results

The hardware test programme (described in Section 2.10.3) produced results consistent with the simulation predictions, with some quantitative differences attributable to real-world hardware behavior not fully captured by the simulation models.

Table 3.1.5 - Hardware Test Results vs. Simulation Expectations

Test	Simulated Expectation	Hardware Result	Deviation & Explanation
EWMA prediction accuracy (HW-01)	RMSE < 5 mAh	RMSE = 1.8 mAh	Better than expected; INA219 noise well-filtered
SOC estimation (HW-02)	Error < 5%	Max error 3.2%	Coulomb counting accurate within 4-hr window
PDR at 100 m / SF7 (HW-04)	~99%	PDR = 99%	Excellent match; no fading at short range
PDR at 2 km / SF9 (HW-05)	~94%	PDR = 94%	Exact match; link budget model valid
PDR at 2 km / SF12 (HW-06)	~99%	PDR = 99%	SF12 fallback confirmed; 3× time-on-air increase
Wave occlusion detection (HW-07)	~95% detection rate	94% detection	1% below model; minor from tank reflections
4-hr ENO test (HW-08)	SOC > 40%	SOC = 55%	Better than expected; actual idle draw lower than modelled
Priority enforcement (HW-03)	HIGH always first	Confirmed	No deviation from algorithm specification

The hardware results closely match simulation expectations across all eight measured test cases, with deviations within acceptable engineering tolerances. The most notable positive deviation is the 4-hour ENO test (HW-08), where the actual SOC after 4 hours of operation was 55% — 15 percentage points higher than the simulation prediction. Investigation revealed that the idle current draw of the actual STM32 hardware (1.8 mA in sleep mode) is lower than the simulation model's value (3.2 mA), which was

based on a conservative datasheet figure. This real-world undercurrent is a beneficial deviation: it implies that actual battery lifetimes will exceed simulation predictions, further supporting ENO compliance in deployment.

3.2 Research Findings

The results presented in Section 3.1 support the following research findings. Each finding is stated as a precise, evidence-grounded conclusion that goes beyond restating the numbers to interpret their significance in the context of the research questions and the broader literature.

3.2.1 Finding 1: ENO and QoS Are Compatible in Marine IoT When Managed by a Cross-Layer Protocol

A central concern in the literature is that Energy Neutral Operation — reducing transmissions to stay within the harvest budget — necessarily degrades Quality of Service, particularly for latency-sensitive high-priority traffic. The ERAC results refute this assumption. In Scenario C (the harshest condition), ERAC simultaneously achieves ENO compliance (100%) and improves HIGH-priority PDR by 37.4 percentage points over the non-adaptive baseline. This is possible because ERAC does not reduce throughput uniformly — it selectively defers low-priority traffic to protect the energy and channel access budget for critical data.

This finding extends the results of Sakib et al. [11], who demonstrated energy-aware QoS MAC for terrestrial WSNs, to the marine context with its additional challenges of stochastic wave-energy harvesting and wave-induced link occlusion. The finding validates the cross-layer design philosophy adopted in ERAC and supports the broader proposition that QoS-aware energy adaptation is preferable to either energy-agnostic MAC (which sacrifices ENO) or pure energy-saving MAC (which sacrifices QoS).

3.2.2 Finding 2: EWMA-Based Energy Prediction Is Sufficient for ENO Enforcement in Solar-Dominant Marine Environments

The EWMA predictor ($\alpha = 0.3$) achieved RMSE of 1.8 mAh in hardware testing and enabled the scheduler to make no incorrect transmission decisions in the sense of transmitting when energy was insufficient for the packet — verified by zero forced battery cutoffs in all 72-hour simulation runs. This challenges the assumption (implicit in some literature) that marine environments require complex predictors such as Kalman filters or machine-learning models to manage stochastic harvesting.

The key insight is that ERAC does not need to predict harvest with high precision; it needs only to predict whether the harvest in the next cycle will be above or below the cost threshold for each priority class. This is a coarser classification task than precise energy forecasting, and EWMA is well-suited to it. The hysteresis bands in the state machine absorb prediction errors of up to ± 5 percentage points in SOC without causing incorrect state transitions, as validated by test HW-02 (max SOC error 3.2%).

This finding has practical significance: it means ERAC can be implemented on microcontrollers with as little as 32 KB of RAM (one order of magnitude below the STM32F401's capacity), making the protocol portable to lower-cost hardware platforms for cost-sensitive deployments.

3.2.3 Finding 3: Wave Occlusion Produces a Distinct Link Degradation Pattern Requiring Asymmetric Adaptation

The fast-drop/slow-rise link quality model ($\alpha_{\text{drop}} = 0.2$, $\alpha_{\text{rise}} = 0.9$) was validated empirically in the water basin test (HW-07), where it correctly identified 94% of wave occlusion events from ACK failure patterns alone, without access to physical wave sensor data. This demonstrates that wave occlusion produces a statistically identifiable signature in the ACK stream — short-duration failures with full recovery — that can be distinguished from permanent link failure using appropriate asymmetric adaptation.

Symmetric adaptation (equal α for drop and rise) was tested during firmware simulation and found to produce excessive retransmission storms as the link quality

metric oscillated with each wave cycle, consuming approximately $2.8\times$ more energy than the asymmetric model in equivalent wave conditions. This finding motivates the specific asymmetric design and contributes a novel protocol mechanism not present in prior marine MAC literature.

3.2.4 Finding 4: Priority-Based Packet Aging Is Essential for Queue Stability Under Sustained Traffic

The comparison between ERAC's two-tier queue with aging and the single-tier FIFO baseline (Table 3.4) reveals that aging is not merely a quality-of-data mechanism — it is operationally essential for queue stability. Without aging, the FIFO queue overflows within 2.3 hours of a sustained traffic burst, causing HIGH-priority packets to be dropped at the enqueue stage — a catastrophic failure mode for a debris-monitoring system. ERAC's aging policy prevents this by continuously reclaiming queue space from stale LOW and MEDIUM packets, ensuring that HIGH-priority packets always have available capacity.

The 18 MEDIUM and 47 LOW aging drops over 72 hours represent less than 1.2% and 2.8% of generated MEDIUM and LOW packets respectively — a negligible freshness-for-availability trade-off given that these packet types carry periodic sensor data with no hard delivery deadline.

3.2.5 Finding 5: Hardware Performance Meets or Exceeds Simulation Predictions

Across all eight hardware test cases, results either matched or exceeded simulation expectations. The most significant exceedance — SOC 15 percentage points above prediction after 4 hours — indicates that the simulation model is conservative with respect to idle power consumption. This conservatism is intentional from an engineering design-safety perspective: a protocol that meets its targets in simulation but exceeds them in hardware is preferable to the reverse. The consistency between simulation and hardware results validates NS-3 as an appropriate evaluation platform

for ERAC protocol development and gives confidence that full-scale deployment performance will be at least as good as simulated.

3.3 Discussion

3.3.1 Achievement of Research Objectives

Chapter 1 stated four specific research objectives. Table 3.3.1 evaluates the degree to which each objective was achieved based on the results and findings presented above.

Table 3.3.1 - Research Objective Achievement Assessment

Objective	Description	Achievement Evidence	Verdict
RO1 — Energy Prediction & Duty Cycling	Implement EWMA energy forecasting; adapt duty cycle to achieve ENO	ENO compliance 100% in all scenarios; EWMA RMSE 1.8 mAh in hardware; 67% network lifetime extension	Fully achieved
RO2 — Priority Scheduling	Multi-level queues with HIGH always served; fairness via heartbeat	HIGH PDR 96.1–99.2%; heartbeat confirmed in 72 hrs; no HIGH packet lost in any run	Fully achieved
RO3 — Link- Aware Operation	Detect wave occlusion; buffer during outage; resume on recovery	94% occlusion detection in HW-07; retransmission on recovery confirmed in INT-03	Substantially achieved (6% missed detections)

RO4 — Cross- Layer Integration	Share energy state with routing; MAC duty pattern informs decisions	Cross-layer SF and power adaptation validated; Raspberry Pi gating confirmed	Achieved within single-hop topology (multi-hop deferred to future work)
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3.3.2 Hypothesis Evaluation

Three hypotheses were stated in Section 1.5. Each is now evaluated against the evidence:

H1: Network lifetime extended by >50% through predictive duty-cycling

Evidence: Network lifetime extended by >67% in Scenario C (baseline 48.2 hrs → ERAC >80 hrs). ENO compliance 100% vs. 61.7% for baseline. CONFIRMED — margin exceeds target by 17 pp.

H2: HIGH-priority latency reduced by at least 20% vs. baseline

Evidence: HIGH-priority latency reduced by 33–77% depending on scenario (Scenario C: 12.6 s → 2.9 s, −77%). CONFIRMED across all scenarios — minimum gain (Scenario A, −33%) comfortably exceeds the 20% target.

H3: HIGH-priority delivery ratio improved in rough conditions vs. baseline

Evidence: HIGH PDR improved by +37.4 pp in Scenario C (storm). Queue overflow preventing HIGH loss eliminated entirely. CONFIRMED — the magnitude of improvement far exceeds what could be achieved by retransmission alone, demonstrating the fundamental contribution of energy-adaptive scheduling.

All three hypotheses are confirmed by the evidence. The degree of confirmation is particularly strong for H3, where the improvement in storm conditions represents an operationally transformative difference: a baseline PDR of 58.7% means that nearly half of all emergency debris detection alerts are lost; ERAC's 96.1% PDR means fewer than 1 in 25 are lost.

3.3.3 Limitations of the Study

The study has several limitations that qualify its conclusions and motivate future work:

1. **Single-hop topology only:** All simulation and hardware testing assumed a single-hop LoRa star topology. Real deployments in larger ocean areas will require multi-hop relay capability. ERAC's cross-layer design is architected for extension to multi-hop routing, but this extension has not been implemented or validated. Results from the current study cannot be directly extrapolated to multi-hop networks.
2. **Synthetic harvest profiles in NS-3:** The simulation uses a mathematical model of solar and wave harvest rather than empirical field data. Real ocean harvest profiles from locations such as the Sri Lankan southern coast, where monsoon weather creates prolonged low-harvest periods, may differ from the sinusoidal model used. Field validation with real harvest data is required before deployment confidence can be established for specific geographic locations.
3. **Small node count in simulation:** Only five buoy nodes were simulated due to computational constraints. In larger networks (20–50 nodes), channel contention effects on the shared LoRa frequency would be more pronounced, and ERAC's probabilistic backoff mechanism may require parameter retuning. The collision rate of <2% observed at 5 nodes may increase non-linearly with node count.

4. **Short hardware test duration:** The longest physical hardware test was 4 hours (HW-08). Long-duration effects such as battery calendar ageing, INA219 drift over temperature cycles, and cumulative Coulomb counting error over days-to-weeks timescales have not been characterised. These effects could impact SOC estimation accuracy and, consequently, state machine behaviour in extended deployment.
5. **Wave energy harvesting not integrated in hardware prototype:** The hardware prototype used a solar panel only; the wave energy harvester described in the system architecture was not physically integrated due to time and resource constraints. The wave harvest contribution was simulated using the IMU-based estimation model (Section 2.5.2), which was calibrated only in tank conditions. Real open-ocean wave energy contribution may differ from tank calibration results.

3.3.4 Comparison with Related Work

Situating ERAC's results against the related work reviewed in Chapter 1 reveals several points of comparison:

Relative to Sakib et al.'s EQPD-MAC [11], which achieved approximately 30% energy reduction in terrestrial WSNs, ERAC achieves an average 38% daily energy consumption reduction in Scenario B and 48% in Scenario C relative to the baseline. The higher reduction is partly attributable to the more extreme energy constraints of the marine harvest profile (which forces deeper adaptation) and partly to ERAC's RPi power gating mechanism, which eliminates the dominant idle consumption component absent in the EQPD-MAC evaluation scenario.

The 96.1% HIGH-priority PDR under storm conditions compares favorably with Liu et al.'s FCLoRa [6], which targets $PRR \geq 88\%$ under moderate wave conditions using physical-layer signal accumulation rather than MAC-layer adaptation. ERAC achieves a higher PDR target through a complementary mechanism — scheduling protection

rather than receiver processing gain — suggesting that combining both approaches in a future protocol could yield further improvement.

The EWMA prediction approach aligns with Vigorito et al.'s ENO-based adaptive duty cycle work [9]. Whereas Vigorito et al. operate in a purely terrestrial context with stable links, ERAC's extension to wave-occlusion-affected channels and the asymmetric link quality model represent novel contributions not present in that foundational work.

3.3.5 Implications for Marine IoT System Design

The results and findings of this study have several practical implications for the design of future marine IoT monitoring systems:

- **Priority-aware MAC is a prerequisite, not an optimization:** Systems that use generic LoRaWAN class A without priority differentiation will, under storm conditions, deliver fewer than 60% of emergency alerts. For safety-critical applications such as debris tracking or oil spill detection, this failure rate is unacceptable. ERAC-class protocols should be considered essential infrastructure, not optional enhancements.
- **Dual INA219 monitoring (harvest + load) is worth the hardware cost:** The two-sensor approach that provides independent harvest and consumption measurements enables accurate Coulomb counting and validates the energy model in real time. A single sensor monitoring only battery voltage provides insufficient temporal resolution for EWMA prediction on 1-second tick intervals.
- **Raspberry Pi on-demand gating provides large energy dividends:** The MOSFET-gated co-processor approach reduces the Raspberry Pi's effective duty cycle from 100% to under 2%, recovering approximately 8–10 mAh per hour that would otherwise be consumed by idle co-processor operation. This saving alone covers the entire MEDIUM-priority transmission budget in Scenario C,

demonstrating the outsized impact of peripheral power management relative to radio duty cycle optimization.

- **ERAC is architecture-agnostic:** Although designed for LoRa, the ERAC scheduling algorithm and state machine are independent of the radio technology. The same protocol logic could be ported to other LPWAN technologies (Sigfox, NB-IoT) or to cellular backhaul for gateway nodes with minimal modification to the radio driver module.

Conclusion

This chapter synthesizes the overall findings of the research, restates the key contributions in the context of the broader field, identifies the limitations of the work, and proposes concrete directions for future research. The chapter concludes with a final reflective statement on the significance of the ERAC protocol for the emerging field of marine IoT.

4.1 Summary of the Research

This dissertation presented the design, implementation, simulation, and physical validation of the Energy-Resilient Adaptive Communication (ERAC) protocol — a cross-layer MAC framework for energy-harvesting IoT buoy networks in marine environments. The research was motivated by a clear and previously unaddressed gap in the literature: no existing MAC protocol simultaneously manages stochastic renewable energy harvesting, intermittent LoRa radio connectivity caused by wave occlusion, and multi-priority Quality of Service requirements, within a single coherent cross-layer design suitable for resource-constrained microcontroller deployment.

The research proceeded through five structured phases. First, a systematic literature review (Chapter 1) identified the specific technical gap and established hypotheses for ERAC's expected performance improvements. Second, the ERAC protocol was formally specified, covering the Energy Neutral Operation constraint model, the three-state energy-aware state machine, the two-tier priority queue architecture, and the `ERAC_SchedulerTick()` scheduling algorithm (Chapter 2, Sections 2.3–2.4). Third, supporting mechanisms were designed including the EWMA-based hybrid energy forecaster, adaptive duty cycling, the fast-drop/slow-rise link quality tracker, adaptive LoRa parameter selection, and the exponential backoff congestion controller (Chapter 2, Sections 2.5–2.6). Fourth, the protocol was implemented as production-quality Embedded C firmware on an STM32F401-based hardware prototype running FreeRTOS, and validated through NS-3 network simulation across three representative marine operating scenarios (Chapter 2, Sections 2.7–2.10). Fifth, results were

presented and analysed (Chapter 3), confirming all three research hypotheses and producing five substantive research findings.

The central results of the research are: ERAC maintains 96.1–99.2% HIGH-priority PDR across all operating scenarios including storm conditions; reduces HIGH-priority end-to-end latency by 33–77% versus the non-adaptive baseline; extends network lifetime by more than 67%; and achieves 100% ENO compliance across all five simulated nodes over 72-hour evaluation periods. All results were confirmed to be consistent between NS-3 simulation and physical hardware testing.

4.2 Key Contributions of the Research

This research makes the following original contributions to the fields of marine IoT, energy-harvesting sensor networks, and MAC protocol design:

1. **ERAC Protocol Specification:** A complete, formally specified cross-layer MAC protocol for marine IoT buoy networks. ERAC is the first protocol in the literature to jointly address stochastic wave-and-solar energy harvesting, wave-induced LoRa link occlusion, and three-tier QoS priority scheduling within a single lightweight MAC design suitable for microcontroller deployment. The protocol is fully documented and reproducible.
2. **Asymmetric Link Quality Model for Wave Occlusion:** The fast-drop/slow-rise link quality tracker (Section 2.6.2) provides a novel mechanism for distinguishing wave-induced transient link failures from permanent outages, using ACK feedback alone without additional physical-layer sensors. The asymmetric parameters ($\alpha_{\text{drop}} = 0.2$, $\alpha_{\text{rise}} = 0.9$) were empirically validated and provide a design template reusable in other marine MAC protocols.
3. **Empirical Validation of EWMA Sufficiency for Marine ENO:** This work provides evidence (Section 3.2.2) that the lightweight EWMA predictor is sufficient for ENO enforcement in solar-dominant marine harvest environments, challenging the implicit assumption in some literature that

marine stochasticity requires complex predictors. This has practical significance for resource-constrained embedded deployment.

4. **Open Marine MAC Simulation Framework:** The NS-3 simulation implementation — including the marine path-loss channel model, wave occlusion Bernoulli outage model, stochastic hybrid harvest model, and ERAC MAC module — constitutes a reusable evaluation framework for future marine MAC protocol research. This framework fills a gap identified in the literature: no standardised marine MAC simulation environment previously existed in the NS-3 ecosystem.
5. **Hardware-Validated Prototype:** The STM32-based hardware prototype with INA219 dual-sensor energy monitoring, SX1276 LoRa communication, RPi co-processor power gating, and FreeRTOS ERAC firmware provides a reproducible reference implementation that other researchers can build upon. All firmware source code is structured for portability to other STM32 family members with HAL-layer modification only.

4.3 Recommendations for Future Work

Based on the limitations identified above and the broader opportunities revealed by this research, the following directions are recommended for future investigation:

- **Multi-Hop ERAC Extension**

The most important and impactful extension of this work is the design and evaluation of ERAC in a multi-hop LoRa mesh topology. This would require: (a) a relay node selection algorithm that incorporates ERAC energy state as a routing metric — preferring relays with higher SOC and better link quality — to prevent routing through energy-depleted nodes; (b) a backpressure mechanism that allows downstream nodes to signal upstream when relay buffers are approaching capacity; and (c) re-evaluation of the NS-3 simulation with 10–50 nodes arranged in representative open-ocean multi-hop topologies.

- **Machine Learning Energy Predictor**

While EWMA was shown to be sufficient for the current operating conditions, a recurrent neural network predictor (specifically, a Long Short-Term Memory model trained on historical solar irradiance and wave height data for the specific deployment location and season) could provide superior prediction accuracy under prolonged cloudy periods and monsoon conditions. Such a model could be trained offline and deployed as a fixed-parameter inference model on the Raspberry Pi co-processor, avoiding the need to run it on the STM32. The energy cost of running periodic inference on the RPi would need to be included in the ENO budget.

- **Physical Wave Energy Harvester Integration**

Integration of a physical wave energy harvester — such as the omnidirectional TENG/EMG nanogenerator described by Zhang et al. [7] — into the hardware prototype would complete the hybrid energy subsystem design and enable empirical characterisation of the wave harvest contribution across realistic sea states. This would also allow the IMU-based wave harvest estimation model (Section 2.5.2) to be calibrated against actual harvester output data, improving harvest prediction accuracy.

- **Lightweight Security for Priority Authentication**

Implementing a lightweight message authentication code (MAC) on ERAC packet headers — using a cipher appropriate for STM32 hardware acceleration, such as AES-128 in CMAC mode — would close the priority spoofing attack surface identified in Section 4.3. A pre-shared symmetric key architecture (one key per node, distributed at provisioning time) would be adequate for the closed-membership OceanSense network and avoids the overhead of asymmetric cryptography.

- **Field Deployment and Long-Duration Validation**

The most critical next step for the OceanSense system is a field deployment in a real ocean environment — ideally the southern coastal waters of Sri Lanka, where the southwest monsoon provides a natural adversarial test condition for the energy management system. A 30-day pilot deployment of 5 nodes would provide: empirical ENO compliance data under real seasonal harvest conditions; characterization of long-duration Coulomb counting drift; real debris detection events to validate the end-to-end pipeline; and data for refining the NS-3 marine channel model with real link measurements

4.4 Closing Remarks

The oceans cover 71% of the Earth's surface and play a fundamental role in regulating climate, supporting biodiversity, and enabling global commerce. Yet the tools available for real-time, continuous, and cost-effective ocean monitoring remain strikingly inadequate relative to the scale and urgency of the monitoring challenge — from plastic debris accumulation to harmful algal blooms to oil spills. IoT-based autonomous buoy networks, powered by renewable ocean energy and connected through long-range radio, represent a technically credible path toward closing this monitoring gap.

The ERAC protocol addresses one of the core engineering barriers to realizing such networks at scale: the tension between the desire to transmit data reliably and the constraint of surviving on intermittent renewable energy in an environment that simultaneously provides that energy and disrupts the radio link used to transmit its observations. By demonstrating that this tension can be resolved — not by compromising on either energy neutrality or communication reliability, but by intelligent, adaptive, cross-layer scheduling — this research contributes a building block toward the vision of a globally distributed, persistent, autonomous ocean sensing infrastructure.

The ERAC firmware, simulation framework, and hardware prototype produced by this research are designed for reproducibility and extension. It is hoped that future researchers will build upon these foundations to develop marine IoT communication protocols that are more robust, more scalable, and ultimately more capable of serving the urgent monitoring needs of the world's oceans.

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Glossary

Appendices